Computer Vision —— Shape from X

2026/10/27

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Slides from CV course with R. Fergus, and S. Savarese, S. Lazebnik, D. Lowe, P. Viola

Agenda

- Shape from X
  - Photometric Stereo
  - Shape from Shading
  - Shape from Motion
  - Shape from Focus / Defocus

3D shape information recovery

Goal: 3D object recognition

cues：

- Stereo vision
- Photometric vision
- Shape from Shading
- Shape from Motion
- Shape from Contour
- Shape from Texture
- Shape from Focus / Defocus

} X

3D shape information recovery

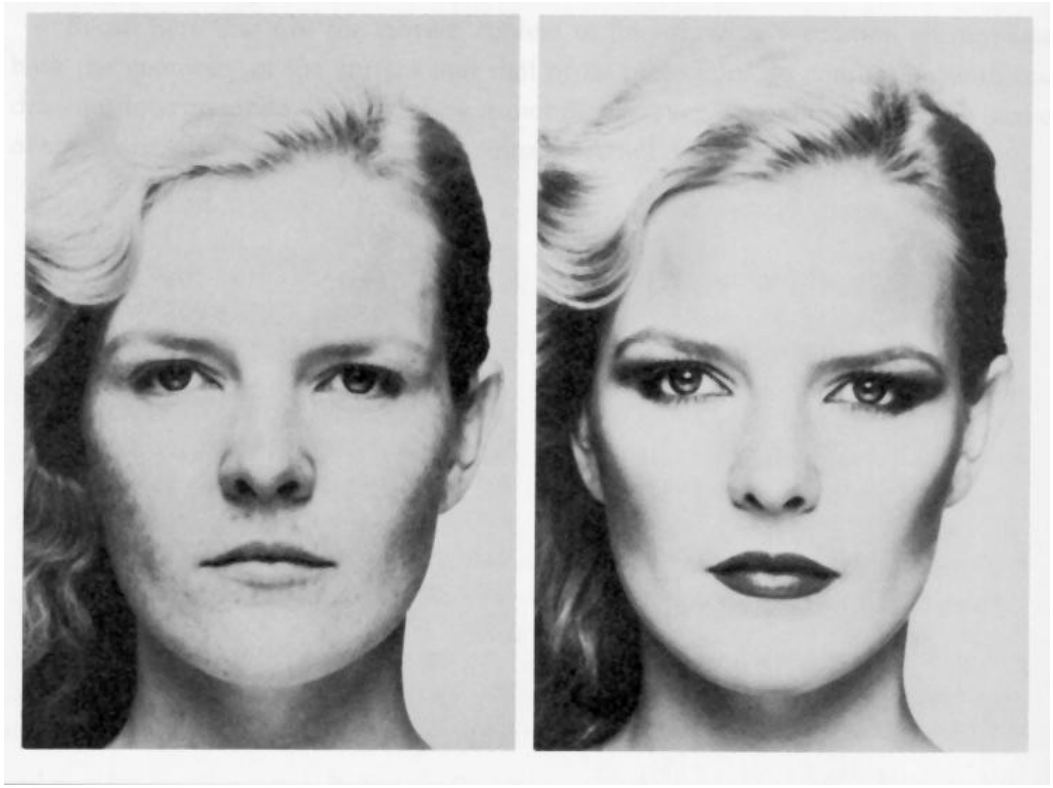
Goal: 3D object recognition

cues：

- Stereo vision
- Photometric vision
- Shape from Shading
- Shape from Motion
- Shape from Contour
- Shape from Texture
- Shape from Focus / Defocus

} X

Photometric Stereo



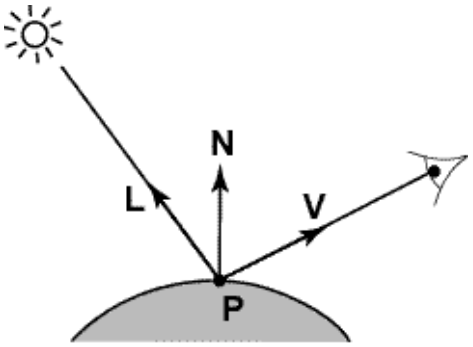
Merle Norman Cosmetics, Los Angeles

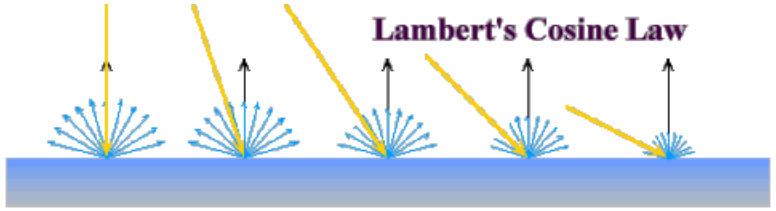
Readings

- R. Woodham, *Photometric Method for Determining Surface Orientation from Multiple Images*. Optical Engineering 19(1)139-144 (1980). (PDF)

Slides by Seitz

Diffuse reflection





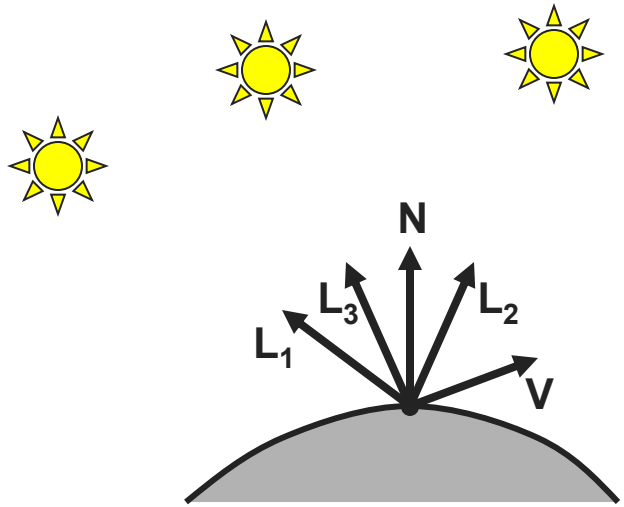
$$R_e = k_d \mathbf{N} \cdot \mathbf{L} R_i$$

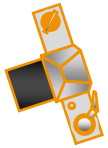
image intensity of **P** → $$I = k_d \mathbf{N} \cdot \mathbf{L}$$

Simplifying assumptions

- I = R_e: camera response function f is the identity function:
  - can always achieve this in practice by solving for f and applying f^{-1} to each pixel in the image
- R_i = 1: light source intensity is 1
  - can achieve this by dividing each pixel in the image by R_i

Photometric stereo





$$I_1 = k_d \mathbf{N} \cdot \mathbf{L_1}$$
$$I_2 = k_d \mathbf{N} \cdot \mathbf{L_2}$$
$$I_3 = k_d \mathbf{N} \cdot \mathbf{L_3}$$

Can write this as a matrix equation:

$$\begin{bmatrix} I_1 & I_2 & I_3 \end{bmatrix} = k_d \mathbf{N}^T \begin{bmatrix} \mathbf{L_1} & \mathbf{L_2} & \mathbf{L_3} \end{bmatrix}$$

Solving the equations

$$\underbrace{\begin{bmatrix} I_1 & I_2 & I_3 \end{bmatrix}}_{\mathbf{I}} = \underbrace{k_d \mathbf{N}^T}_{\mathbf{G}} \underbrace{\begin{bmatrix} \mathbf{L_1} & \mathbf{L_2} & \mathbf{L_3} \end{bmatrix}}_{\mathcal{L}}$$

$\mathbf{I}$: 1×3 $\quad$ $\mathbf{G}$: 1×3 $\quad$ $\mathcal{L}$: 3×3

$$\mathbf{G} = \mathbf{I}\mathbf{L}^{-1}$$

$$k_d = \|\mathbf{G}\|$$

$$\mathbf{N} = \frac{1}{k_d}\mathbf{G}$$

Photometric Stereo

compute surface shape from normal:

line integral

1. suppose the depth of point $p_0(x_0,y_0)$ is $z_0(x_0,y_0)$
2. compute the relative depth of point p(x,y)

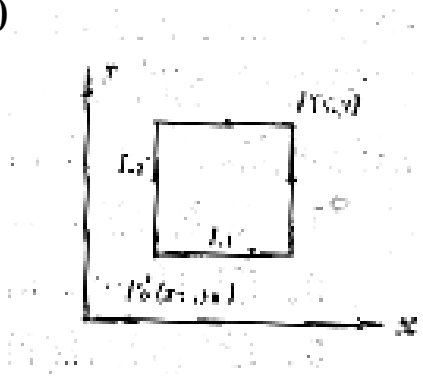
$$z(x,y) = z_0(x_0,y_o) + \sum_{i=1}^{k} w_i \int_{L_i(x_0,y_0)}^{(x,y)} (pdx + qdy)$$

L_i(I=1,2,…,k) is the path of integration from $p_0(x_0,y_0)$ to p(x,y)

ω_i is the weight of every path, and $\sum_{i+1}^{k} \omega_i = 1$, $\omega_i \geq 0$

$p=-n_1/n_3$, $q=-n_2/n_3$, $N=[n_1, n_2, n_3]^T$

Two paths of integration:



$$\omega_1 = \omega_2 = \frac{1}{2}$$

$$z(x,y) = z_0(x_0,y_o) + \frac{1}{2}\int_{L_1(x_0,y_0)}^{(x,y)} (pdx + qdy) + \frac{1}{2}\int_{L_2(x_0,y_0)}^{(x,y)} (pdx + qdy)$$

$$= z_0(x_0,y_o) + \frac{1}{2}\left(\int_{y_0}^{y} q(x_0,y)dy + \int_{x_0}^{x} p(x,y)dx\right) + \frac{1}{2}\left(\int_{x_0}^{x} p(x,y_o)dx + \int_{y_o}^{y} q(x,y)dy\right)$$

Photometric Stereo

compute surface shape from normal:

variation

weakness of line integral: sensitive to noise, dependent on the path of integration

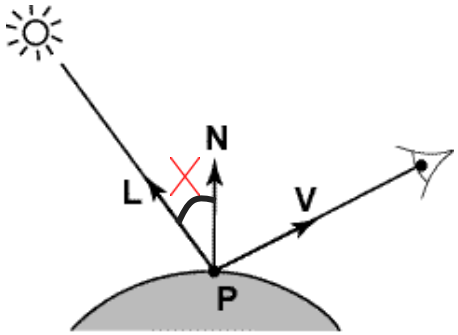
Solution: find the best fitting surface z*

$$z^* = \min_z \iint ((z_x - p)^2 + (z_y - q)^2 \, dxdy$$

Euler equation

$$\nabla^2 z = p_x + q_y$$

Shape from shading



- Suppose

$$k_d = 1$$

$$I = k_d \mathbf{N} \cdot \mathbf{L}$$

$$= \mathbf{N} \cdot \mathbf{L}$$

$$= cos\ \theta_i$$

You can directly measure angle between normal and light source

- Not quite enough information to compute surface shape
- But can be if you add some additional info, for example
  - assume a few of the normals are known (e.g., along silhouette)
  - constraints on neighboring normals—"integrability"
  - smoothness

Shape from shading

Regularization method， minizing E:

$$E = \sum_{x,y} \left(I(x,y) - \mathbf{N} \bullet \mathbf{L} \right)^2 + \lambda \sum_{x,y} F(x,y)$$

Continuity Constraint, change rate of normal:

$$F(x,y) = (p(x+1,y) - p(x,y))^2 + (q(x+1,y) - q(x,y))^2 + (p(x,y+1) - p(x,y))^2 + (q(x,y+1) - q(x,y))^2$$

$p=-n_1/n_3$, $q=-n_2/n_3$, $N=[n_1, n_2, n_3]^T$

$$n = (-p, -q, 1)/\sqrt{1+p^2+q^2}$$

Solution: compute the derivatives of E relative to p and q to 0, and set them to 0.

Refer to K. Ikeuchi, B.K.P.Horn, Numerical shape from shading and occluding boundaries, Artificial Intelligence, 1981, 17:141-184

Shape from shading

Weakness of Regularization method:

Continuity constraint is only suitable for a single object. For an image with multiple objects, image segmentation is needed, and define E for each object. For the edges of objects, uncontinuity should be taken into account.

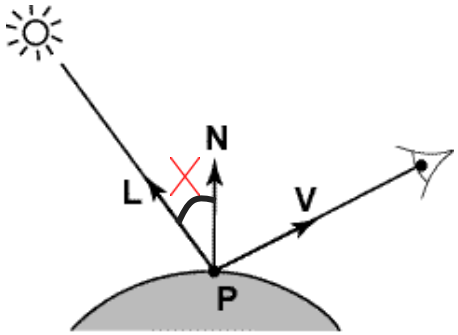
When solving p and q, two order mixed derivative constraint is not considered, i.e., $\frac{\partial z}{\partial x \partial y} = \frac{\partial z}{\partial y \partial x}$

Its discrete form is

$$p(x, y+1) - p(x, y) = q(x+1, y) - q(x, y)$$

Refer to R.M.Haralick, L.G.Shapiro, Computer and Robot Vision, Addison-Wesley Publishing Company, Inc. 1993

Shape from shading



- Suppose

$$
\begin{aligned}
k_d &= 1 \\
I &= k_d \mathbf{N} \cdot \mathbf{L} \\
&= \mathbf{N} \cdot \mathbf{L} \\
&= cos\ \theta_i
\end{aligned}
$$

You can directly measure angle between normal and light source

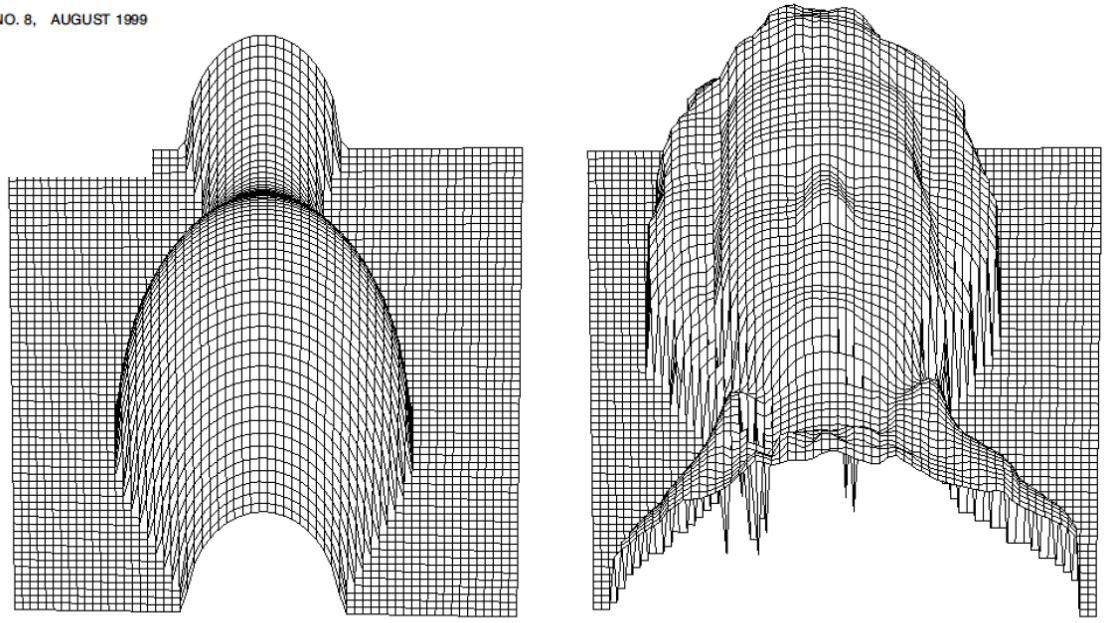
- Not quite enough information to compute surface shape
- But can be if you add some additional info, for example
  - assume a few of the normals are known (e.g., along silhouette)
  - constraints on neighboring normals—"integrability"
  - smoothness
- Hard to get it to work well in practice
  - plus, how many real objects have constant albedo?

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 21, NO. 8, AUGUST 1999

Shape from Shading: A Survey

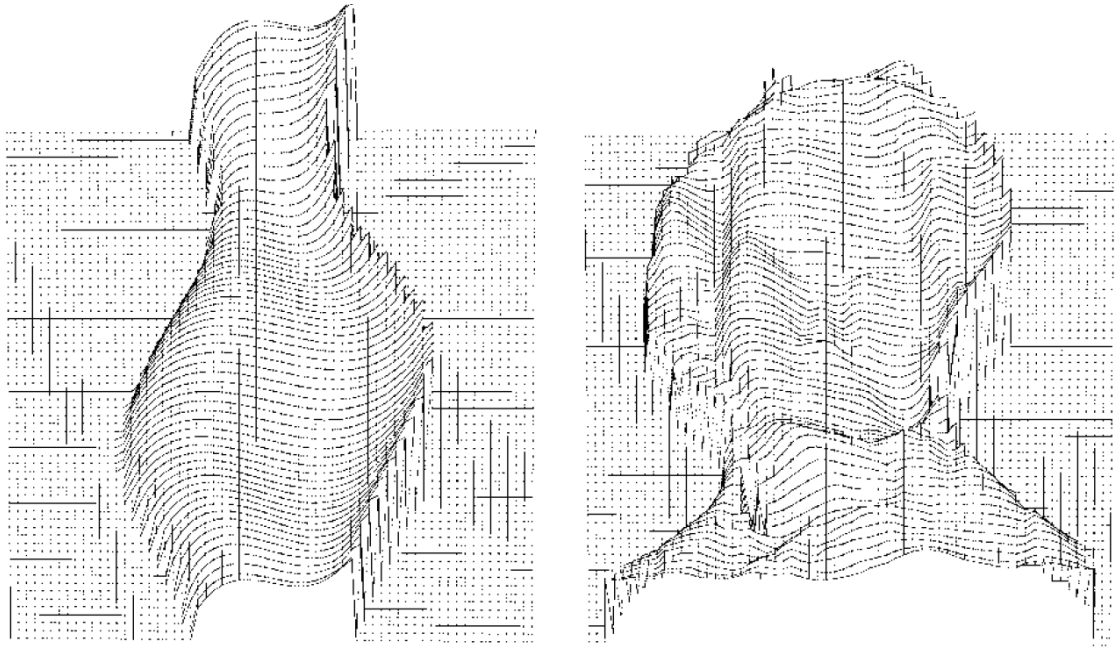
Ruo Zhang, Ping-Sing Tsai, James Edwin Cryer, and Mubarak Shah

Ground truth



(a) (b)

Linear shape from shading



(a) (b)

Shape from motion

- Moving objects: rigid body
- Feature-based: point features
- Steps: 1. point matching; 2. compute 3D motion and object shape from point correspondence
- Method:
  - 1. solve nonlinear equation
  - 2. solve linear equation

Shape from motion - orthogonal projection

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = R\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + T = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \bullet \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}$$

$$X = x, \quad Y = y$$
$$X' = x', \quad Y' = y'$$

- Motion analysis: use N corresponding point pairs from two images $((x_i, y_i),(x_i', y_i')),(i = 1,2,\cdots,N)$, estimate motion parameters R and T, and the 3D parameters of moving object
- 3N equations vs 12+2N unknown numbers: N>=12
- Problem: ΔZ is not necessary (object shape only), Z' is not necessary as well

Shape from motion - orthogonal projection

$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} \Delta X \\ \Delta Y \end{bmatrix}$$

$$((x_i, y_i), (x_i', y_i')), (i = 1, 2, \cdots, N)$$

- For N corresponding point pairs from two images, 2N equations vs 8+N unknown numbers: N>=8
- Problem: absolute coordinates vs relative coordinates

Shape from motion - orthogonal projection

$$\begin{bmatrix} X' - X'_0 \\ Y' - Y'_0 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \end{bmatrix} \begin{bmatrix} X - X_0 \\ Y - Y_0 \\ Z - Z_0 \end{bmatrix}$$

$$((x_i, y_i), (x'_i, y'_i)), (i = 1,2,\cdots, N)$$

- For N corresponding point pairs from two images, 2N equations vs 6+N unknown numbers: N>=6
- Due to the reference point, N>=7

Shape from motion - perspective projection

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + T \qquad x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

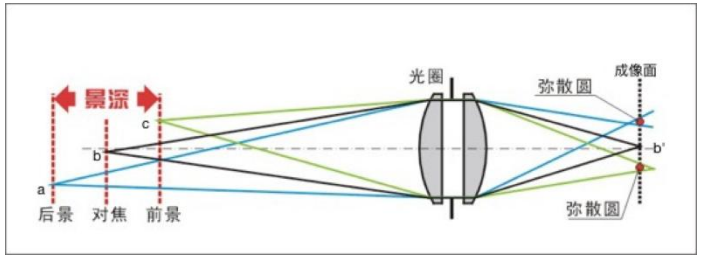
$$x' = f\frac{X'}{Z'} \quad y' = f\frac{Y'}{Z'}$$

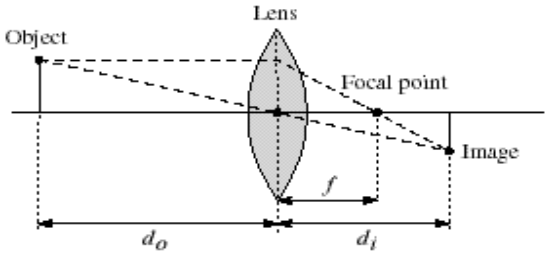
$$Z' \bullet \begin{bmatrix} \frac{1}{f}x' \\ \frac{1}{f}y' \\ 1 \\ \end{bmatrix} = Z \bullet R \begin{bmatrix} \frac{1}{f}x \\ \frac{1}{f}y \\ 1 \\ \end{bmatrix} + T \qquad ((x_i, y_i), (x_i', y_i')), (i = 1,2,\cdots, N)$$

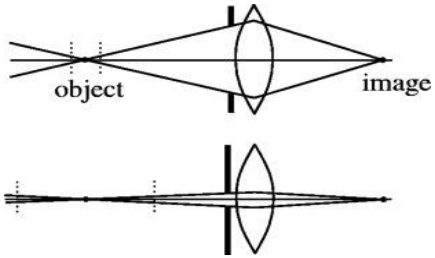
– For N corresponding point pairs from two images, 3N equations vs 12+2N unknown numbers: N>=12

根据焦距确定深度









$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

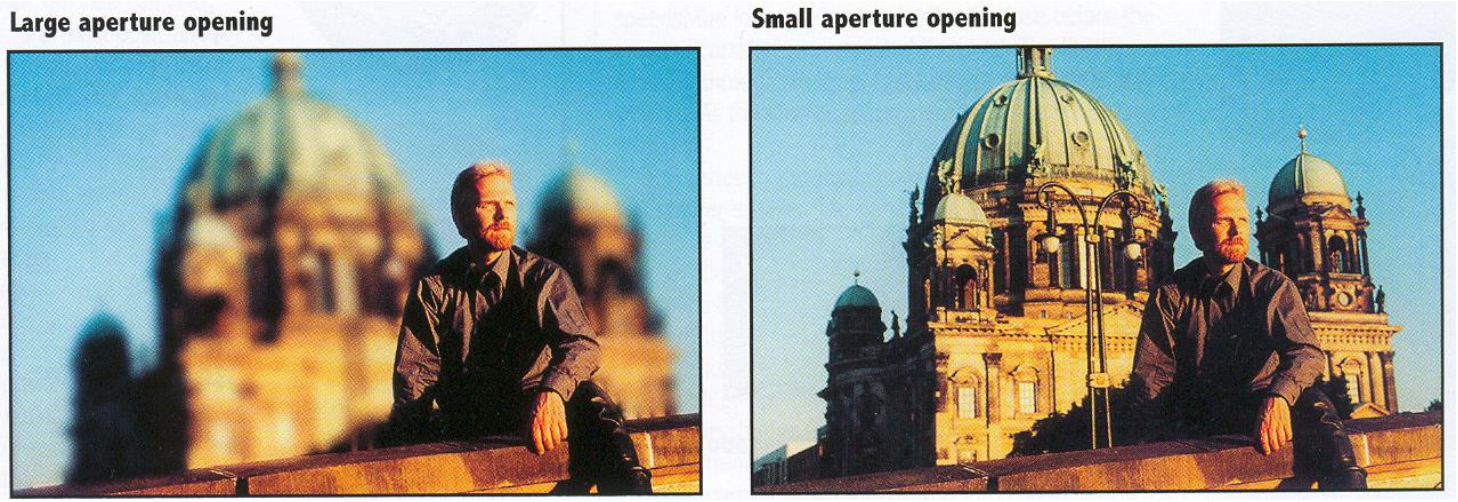


Image and Depth from a Conventional Camera with a Coded Aperture

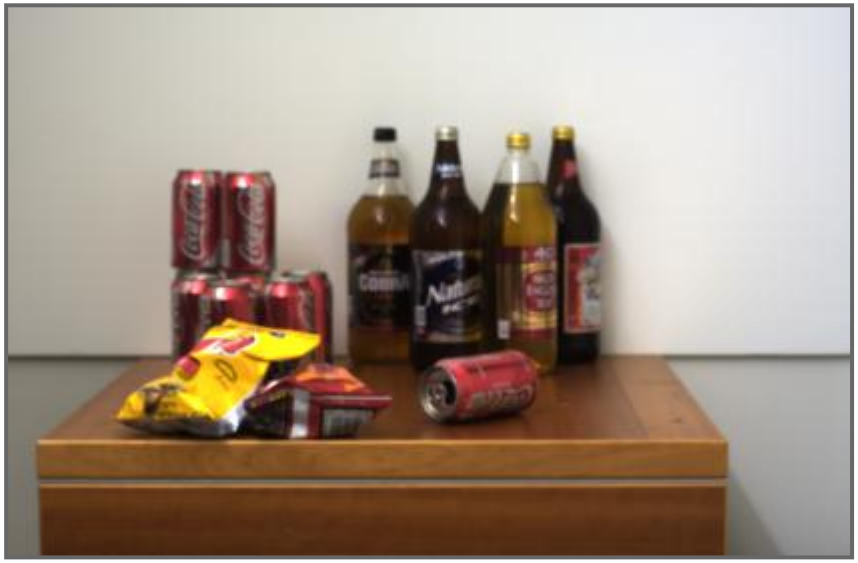
Anat Levin, Rob Fergus, Frédo Durand, William Freeman

MIT CSAIL

Output #1: Depth map



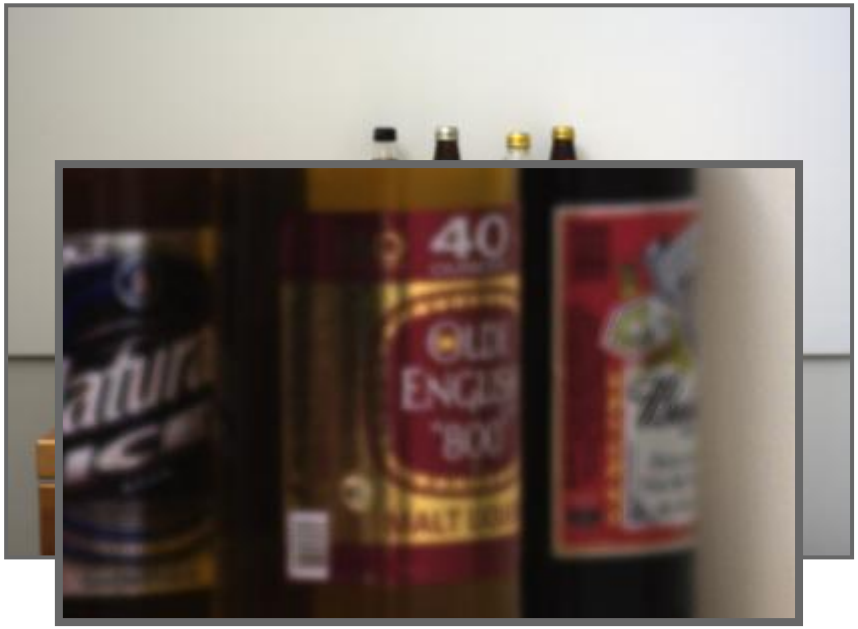
Single input image:



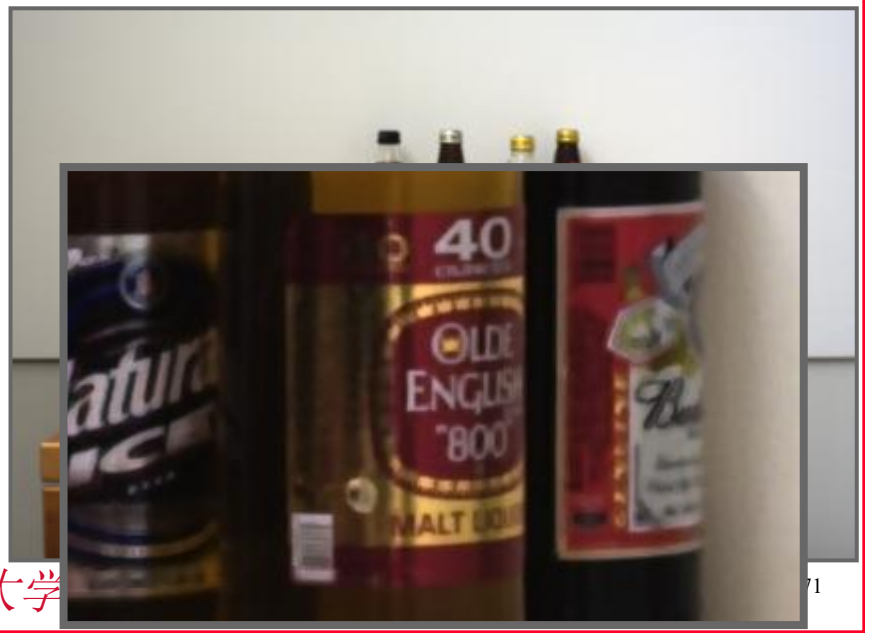
Output #1: Depth map



Single input image:



Output #2: All-focused image



Lens and defocus

Lens’ aperture



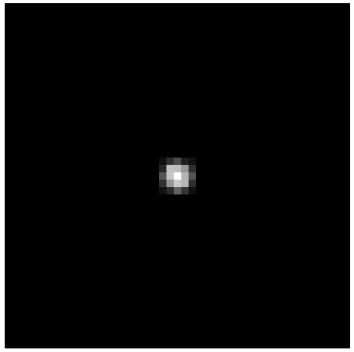
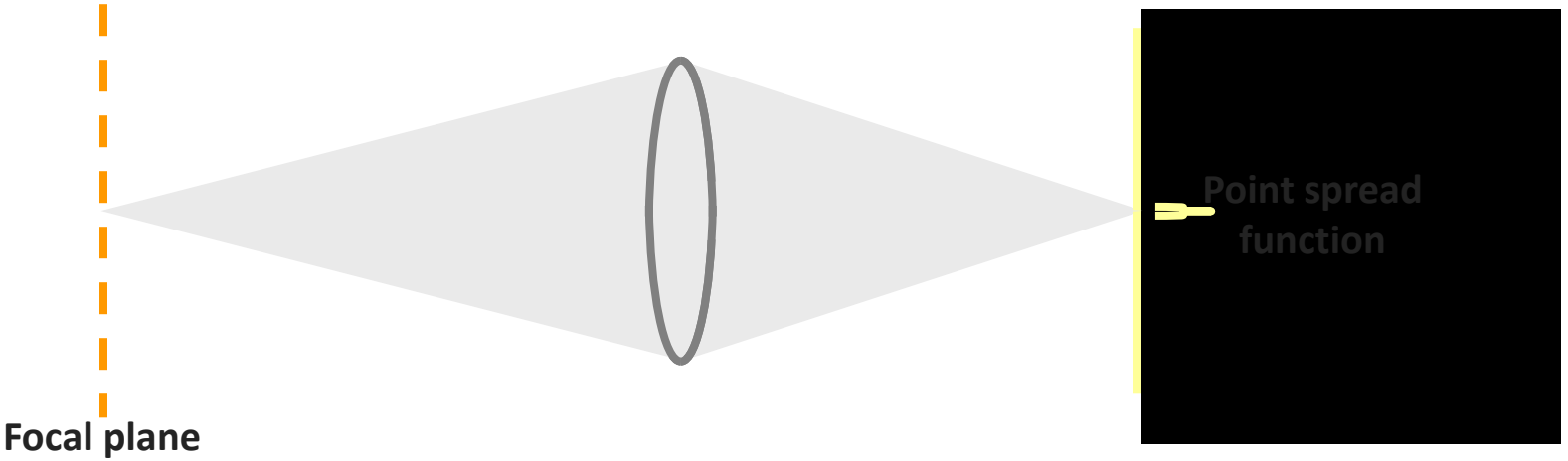


Image of a point light source



Lens and defocus

Lens’ aperture



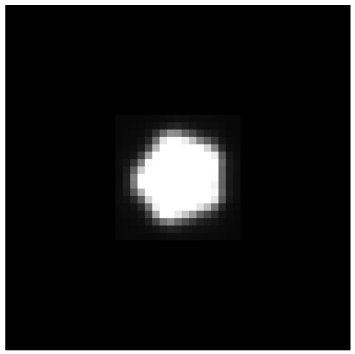
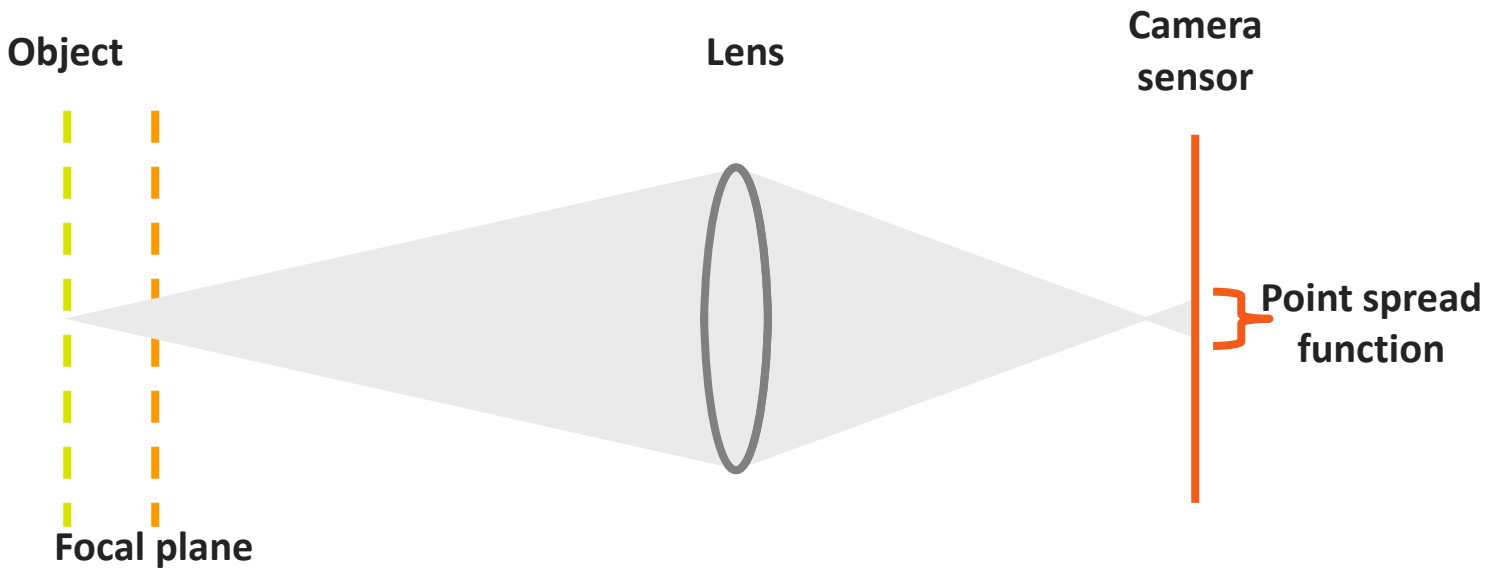


Image of a defocused point light source



Lens and defocus

Lens’ aperture



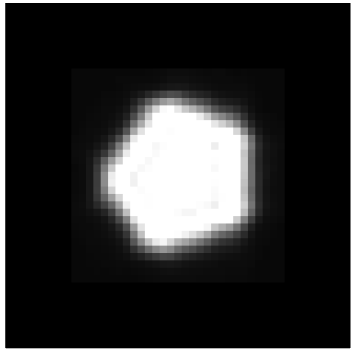
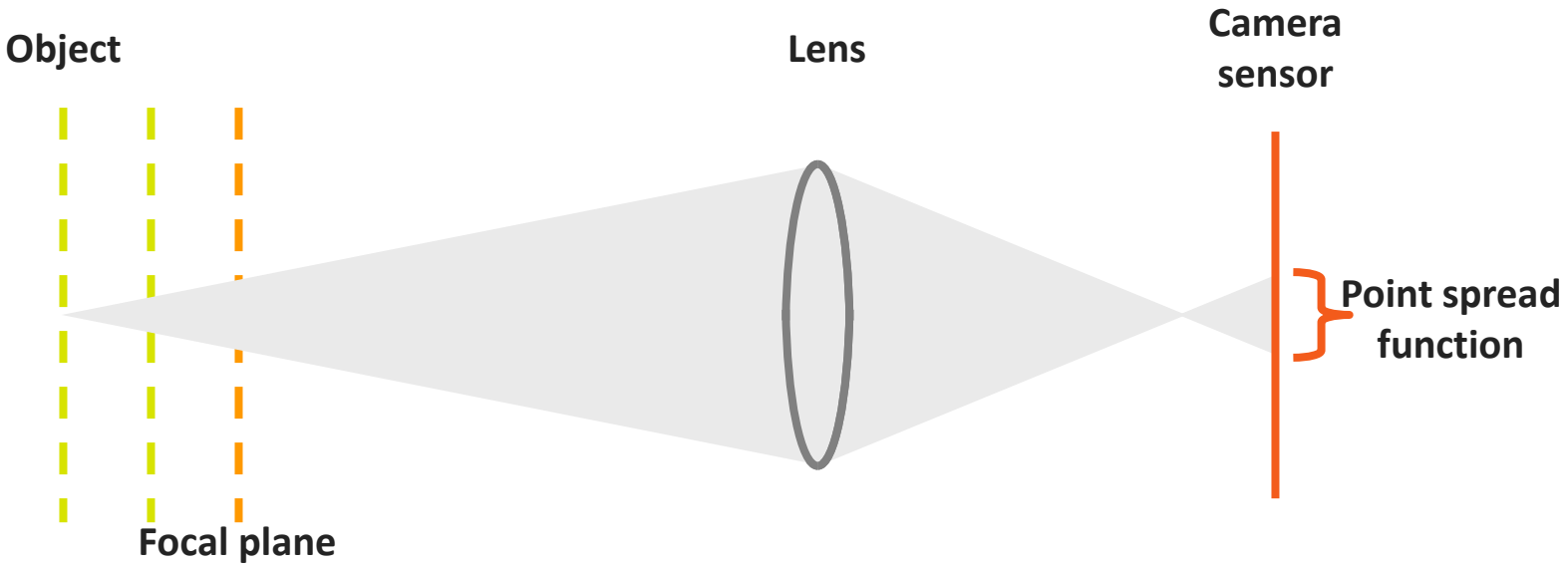


Image of a defocused point light source



Lens and defocus

Lens’ aperture



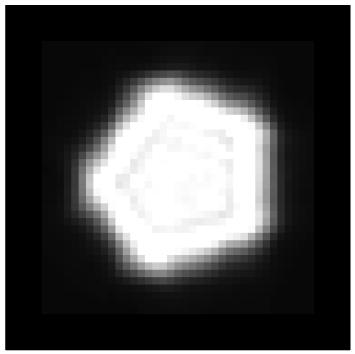
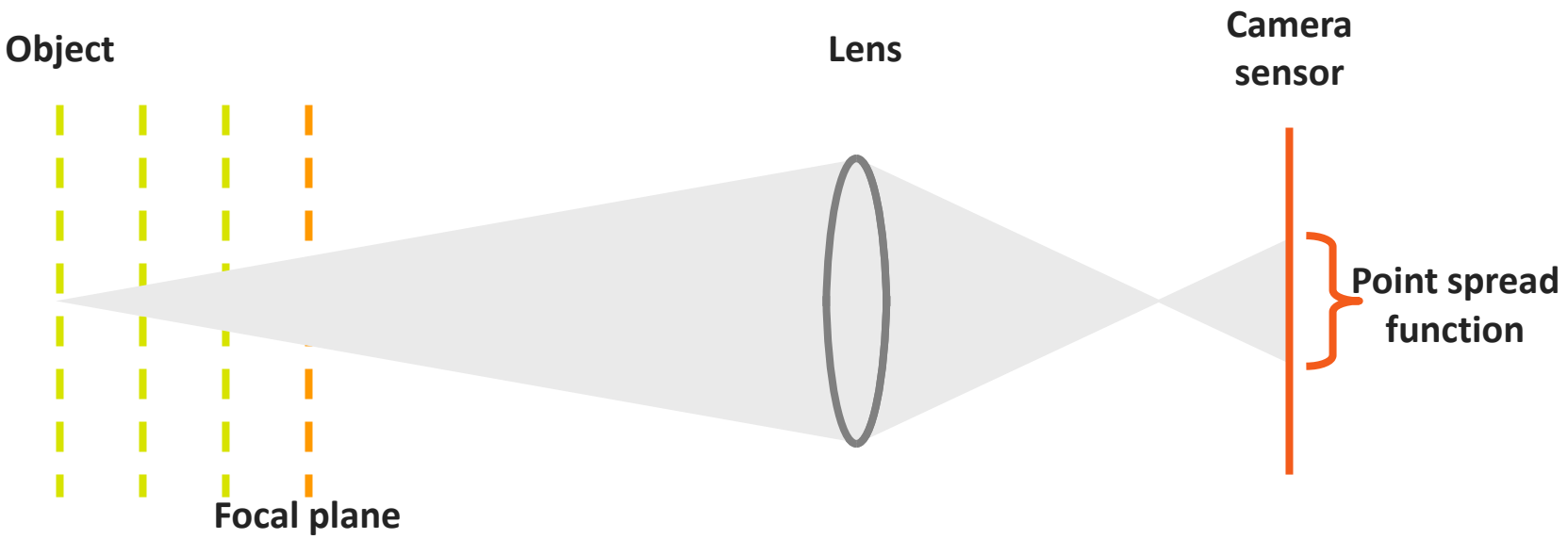


Image of a defocused point light source



Lens and defocus

Lens’ aperture



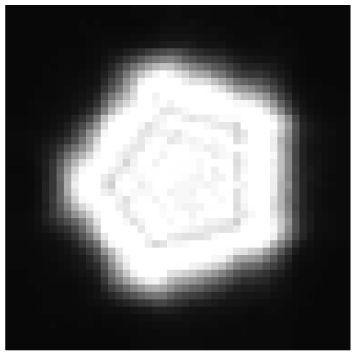
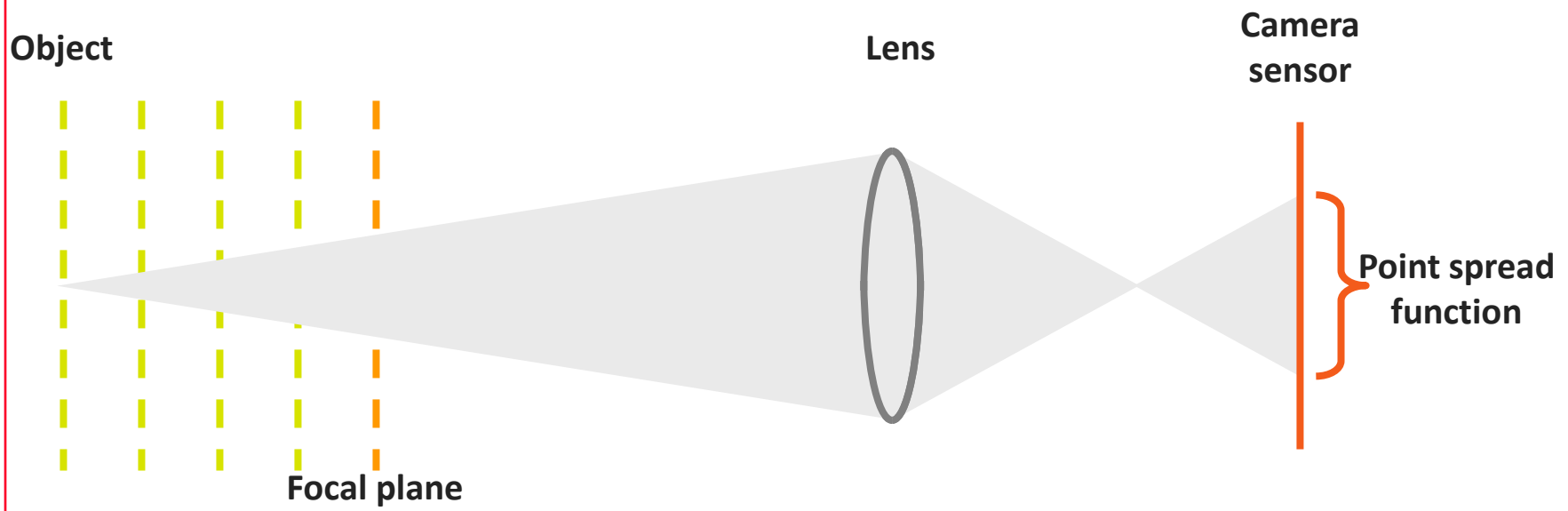


Image of a defocused point light source



Depth and defocus



ill posed

Depth from defocus:

Infer depth by analyzing local scale of defocus blur



Challenges

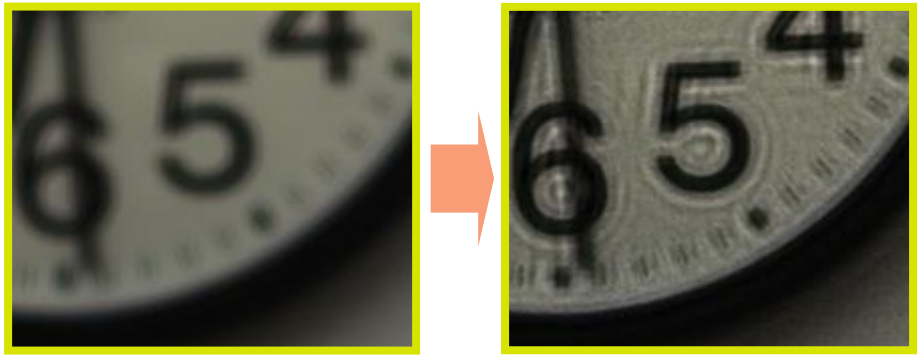
- Hard to discriminate a smooth scene from defocus blur

?

Out of focus



- Hard to undo defocus blur



Input

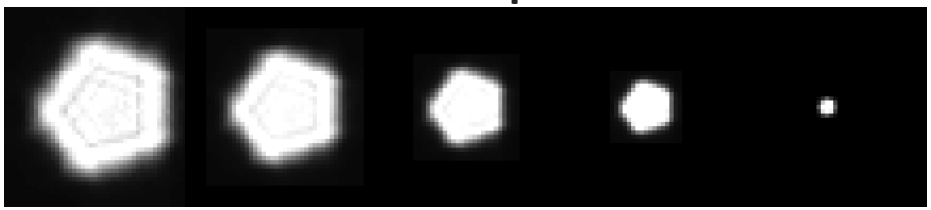
Ringing with conventional deblurring algorithm

Defocus as local convolution

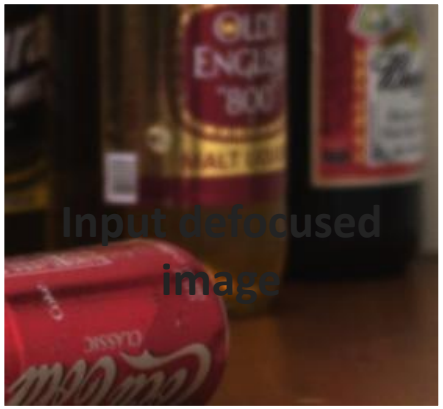


Input defocused image

Calibrated blur kernels at different depths



Defocus as local convolution



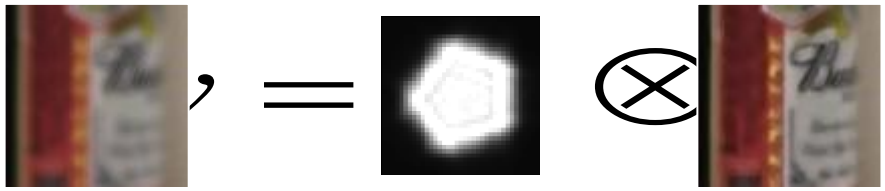
$$y = f_k \square x$$

Local sub-window

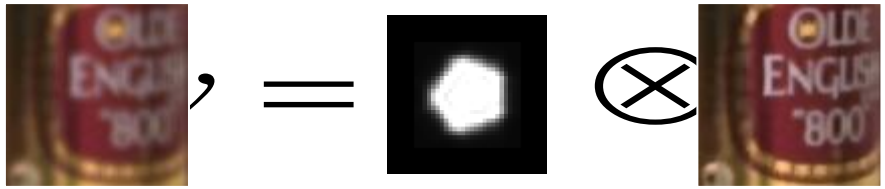
Calibrated blur kernels at depth k

Sharp sub-window

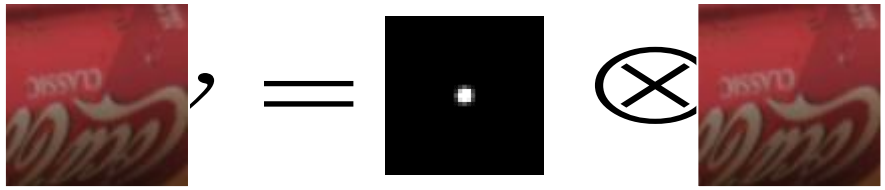
Depth k=1:



Depth k=2:

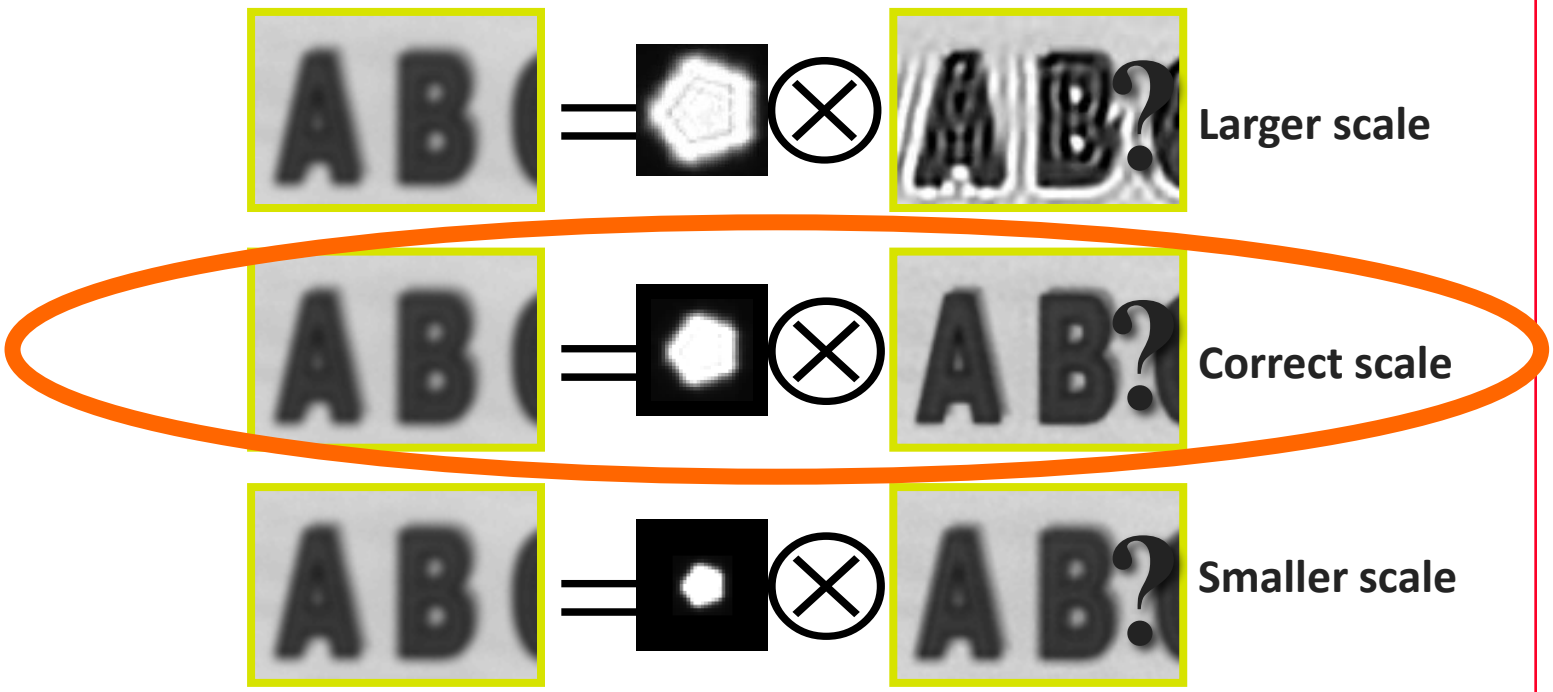


Depth k=3:



Overview

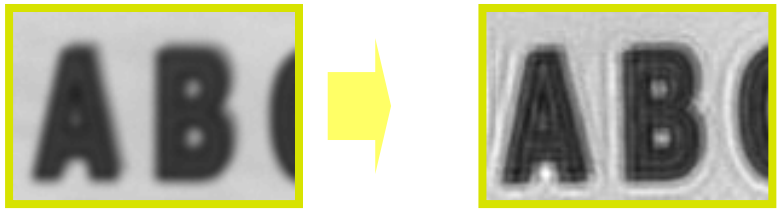
Try deconvolving local input windows with different scaled filters:



Somehow: select best scale.

Challenges

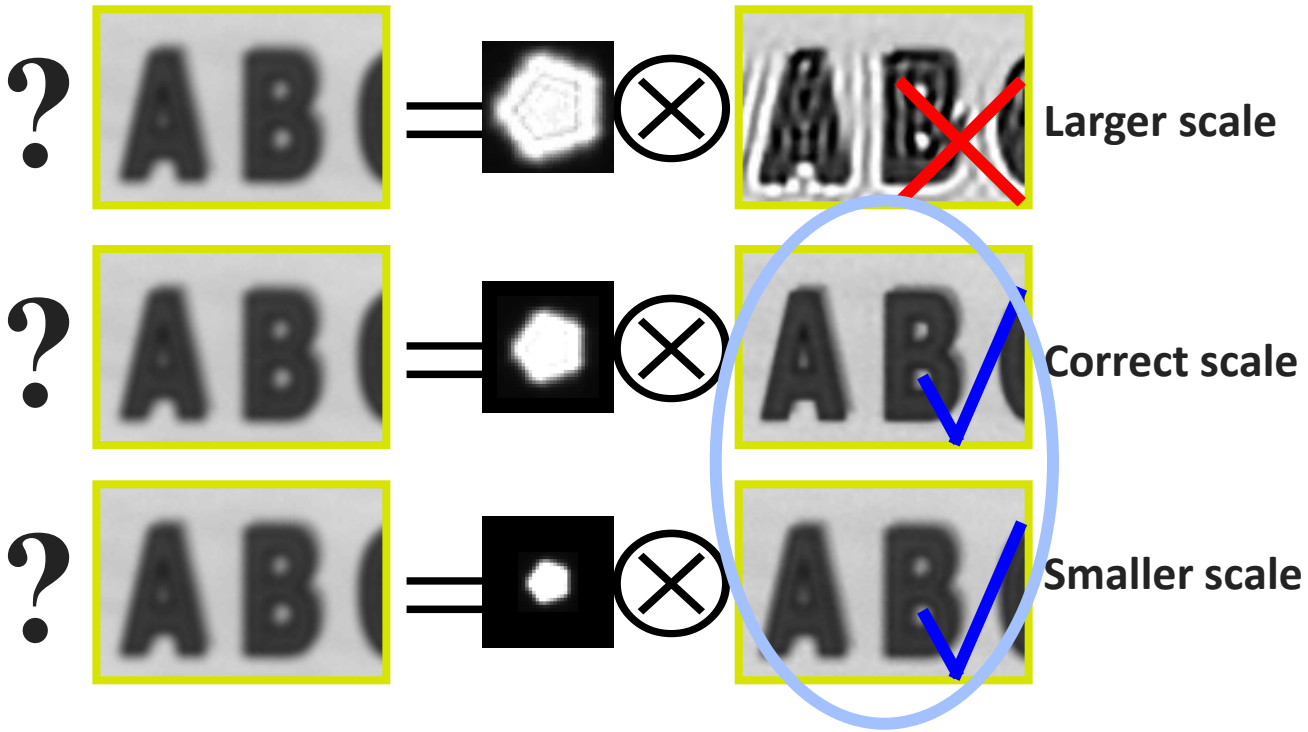
- **Hard to deconvolve even when kernel is known**



Input

Ringing with the traditional Richardson-Lucy deconvolution algorithm

- **Hard to identify correct scale:**



Deconvolution is ill posed

$$f \otimes x = y$$



Deconvolution is ill posed

$$f \otimes x = y$$

Solution 1:

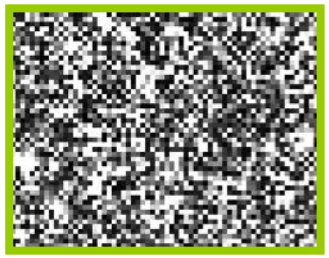


Solution 2:



Idea 1: Natural images prior

What makes images special?

	Natural	Unnatural	
Image			
gradient			

Natural images have sparse gradients

⇒ put a penalty on gradients

Deconvolution with prior

$$x = \arg\min \underbrace{| f \otimes x - y |^2}_{\text{Convolution error}} + \lambda \underbrace{\sum_i \rho(\nabla x_i)}_{\text{Derivatives prior}}$$



Equal convolution error



Comparing deconvolution algorithms



Input

(Non blind) deconvolution code available online:
http://groups.csail.mit.edu/graphics/CodedAperture/

$$\rho(\nabla x) = \|\nabla x\|^2$$

"spread" gradients

$$\rho(\nabla x) = \|\nabla x\|^{0.8}$$

"localizes" gradients



Richardson-Lucy

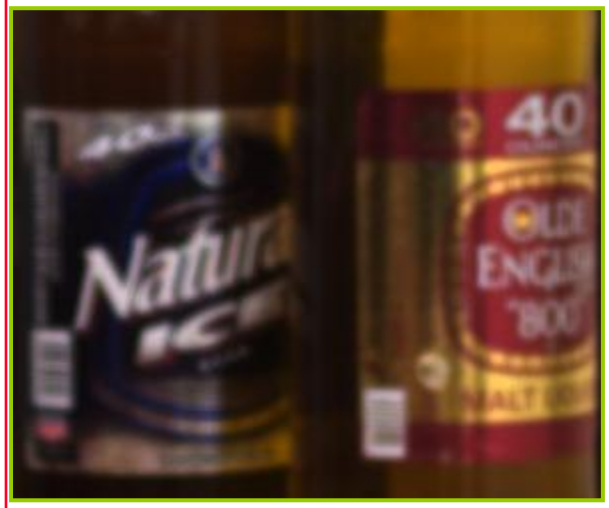


Gaussian prior



Sparse prior

Comparing deconvolution algorithms



Input

(Non blind) deconvolution code available online:
http://groups.csail.mit.edu/graphics/CodedAperture/

$$\rho(\nabla x) = \|\nabla x\|^2$$

"spread" gradients

$$\rho(\nabla x) = \|\nabla x\|^{0.8}$$

"localizes" gradients



Richardson-Lucy



Gaussian prior



Sparse prior

- 基于深度判别式先验的图像盲反卷积

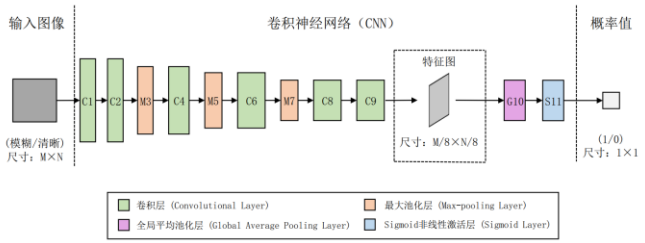
  将训练好的二分类器嵌入至传统的盲反卷积优化框架中，使得迭代过程中潜在图像的中间结果更符合清晰图像的特征

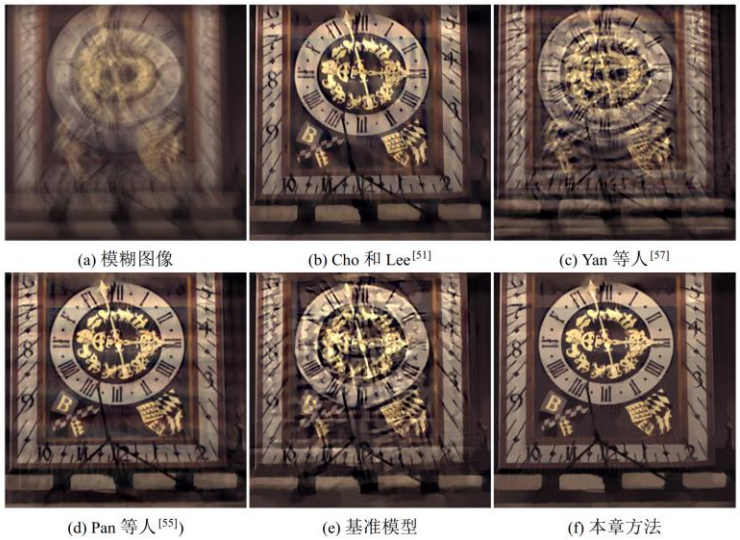
$$(k^*, I^*) = \min_{I,k} \|I \otimes k - B\|_2^2 + \lambda_k p(k) + \lambda_I p(I)$$

数据逼近

模糊核约束

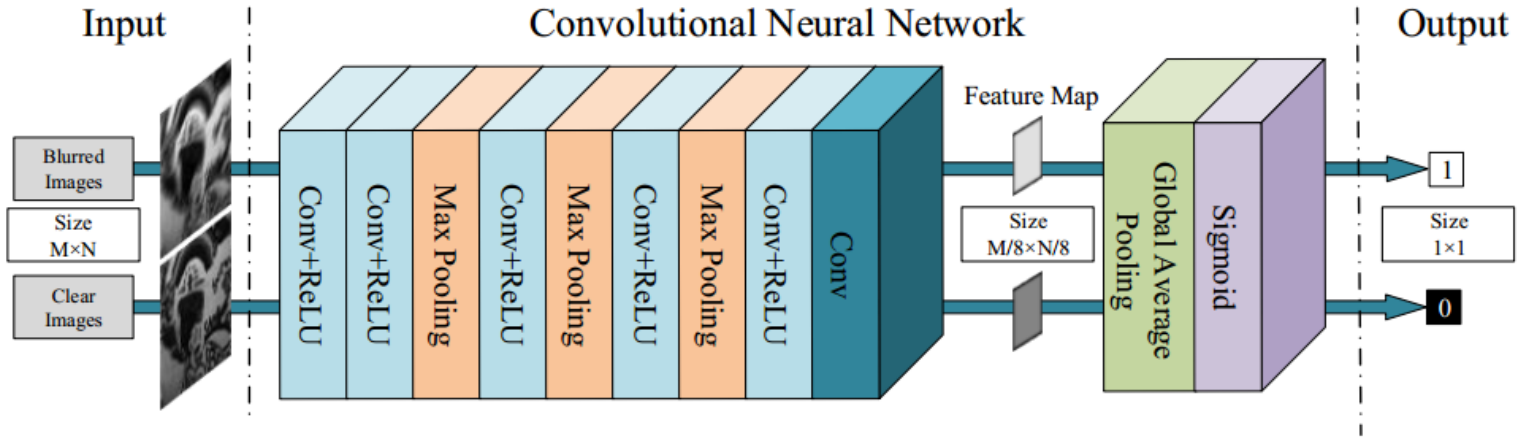
数据驱动的图像先验约束





Learning a Discriminative Prior for Blind Image Deblurring. CVPR 2018
Blind Image Deblurring via Deep Discriminative Priors. IJCV 2019

深度判别式先验



总模型 (4)

$$\min_{I,k} \ \|I \otimes k - B\|_2^2 + \gamma\|k\|_2^2 + \mu\|\nabla I\|_0 + \lambda f(I)$$

L2 数据项 + 核正则 + L0 梯度先验 + CNN 判别先验

- When the image is blurred, the response of the CNN is close to 1.
- When the image is clear, the response of the CNN is close to 0.
- Blurred region, red in the feature map
- Clear region, blue in the feature map

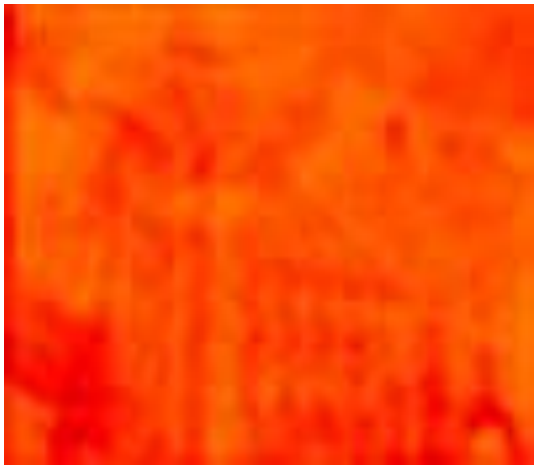
深度判别式先验



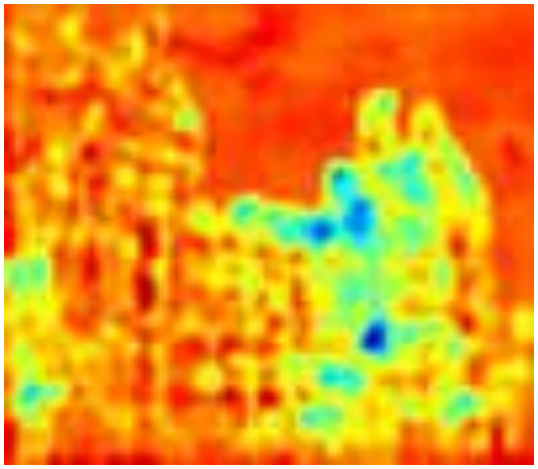
Blurred image (Motion)



Clear image



Blurred feature (Motion)



Clear feature

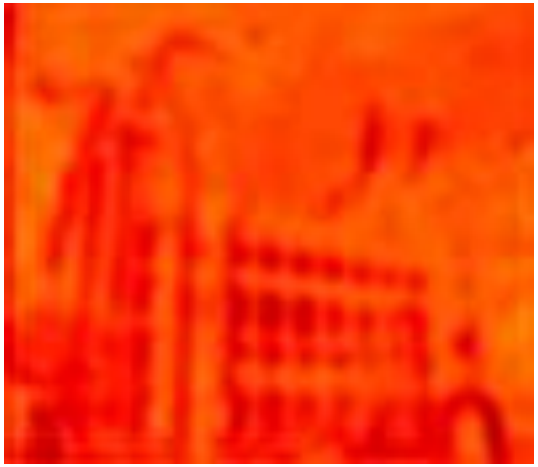
深度判别式先验



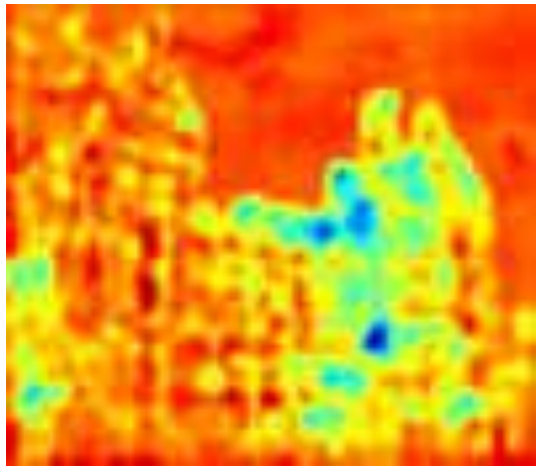
Blurred image (Defocus)



Clear image



Blurred feature (Defocus Motion)



Clear feature

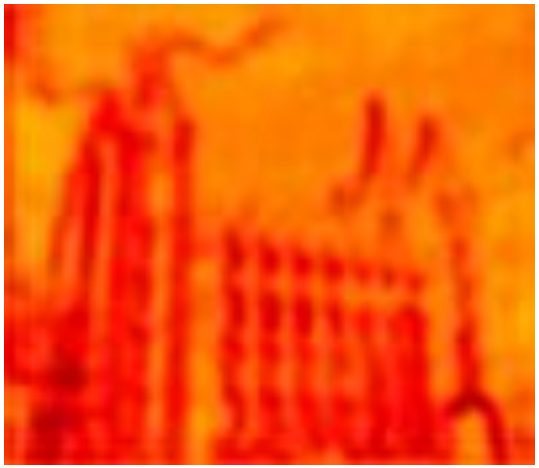
深度判别式先验



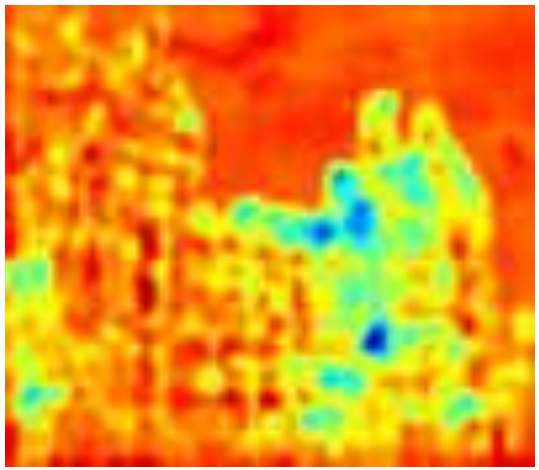
Blurred image (Gaussian)



Clear image



Blurred feature (Gaussian)



Clear feature

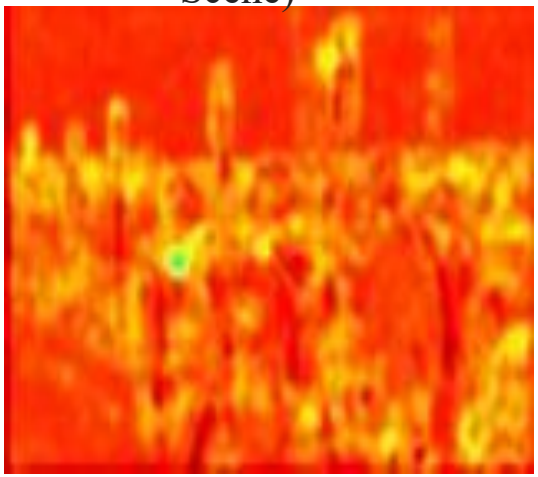
深度判别式先验



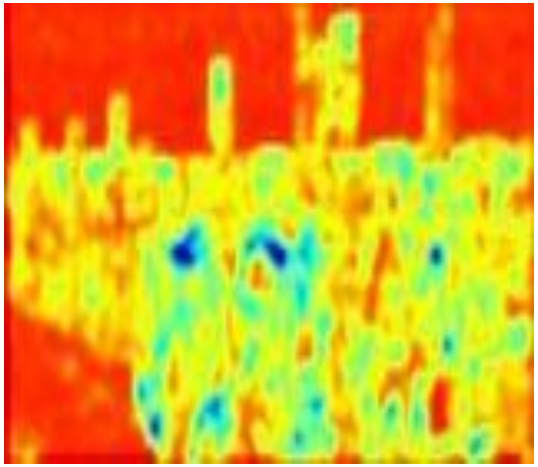
Blurred image (Dynamic Scene)



Clear image



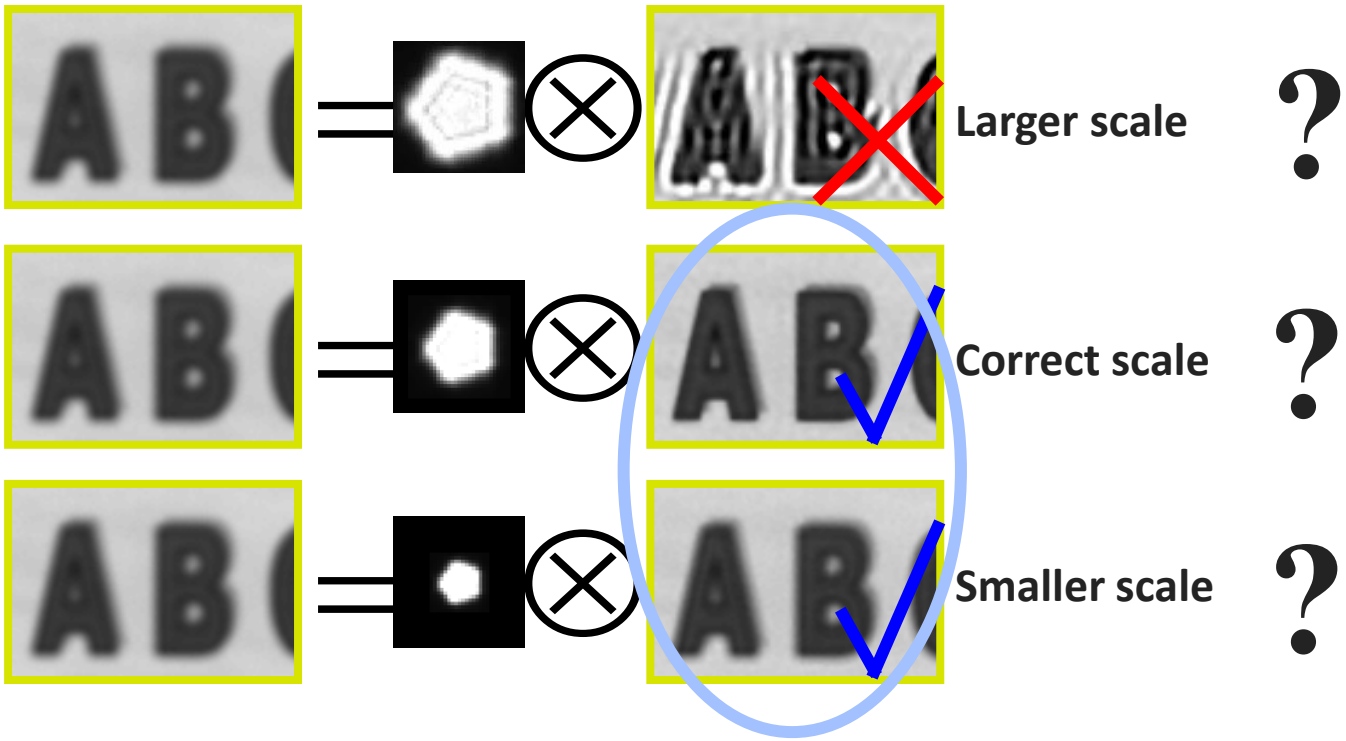
Blurred feature (Dynamic Scene)



Clear feature

Recall: Overview

Try deconvolving local input windows with different scaled filters:



Somehow: select best scale.

Challenge: smaller scale not so different than correct

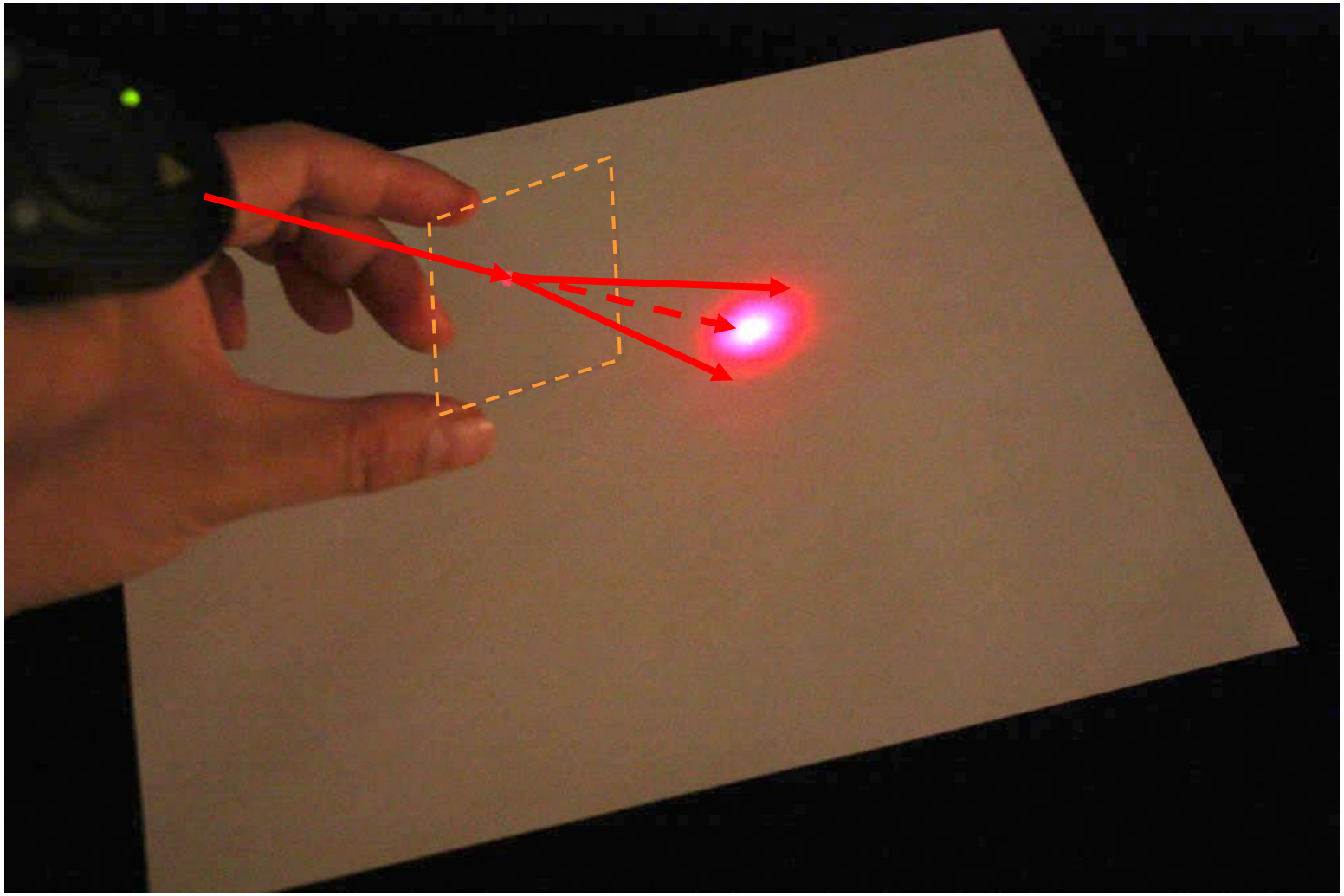
Depth from Diffusion

Changyin Zhou Oliver Cossairt Shree Nayar

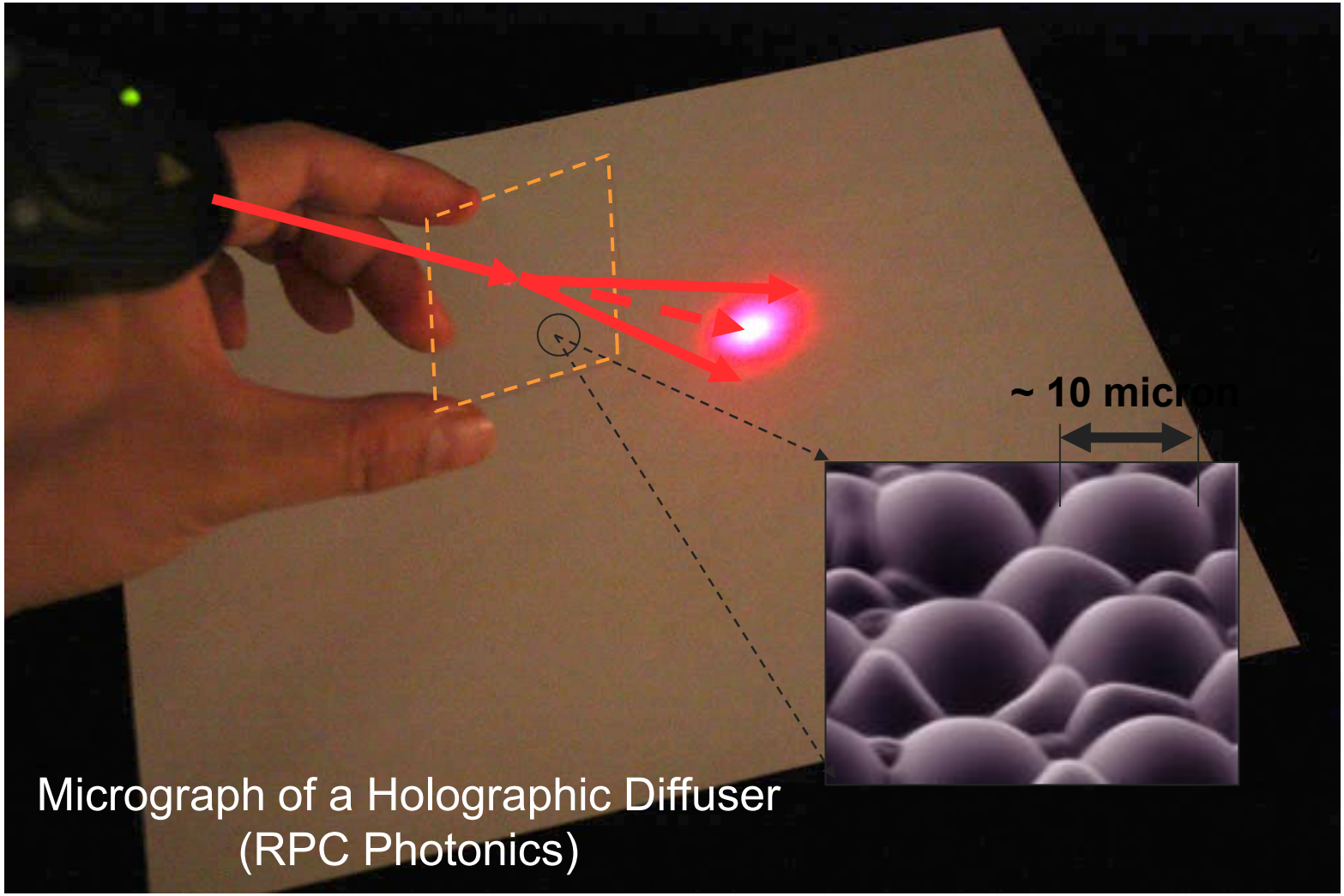
Columbia University

Supported by ONR

Optical Diffuser



Optical Diffuser

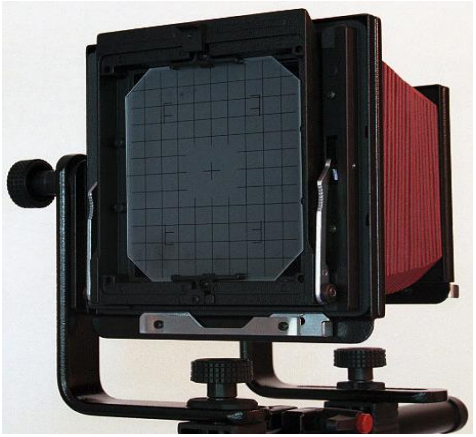


[Gray, 1978] [Chang et al., 2006] [Garcia-Guerrero et al. 2007]

Diffusers as Accessories



(B&H)
Diffusers for illumination

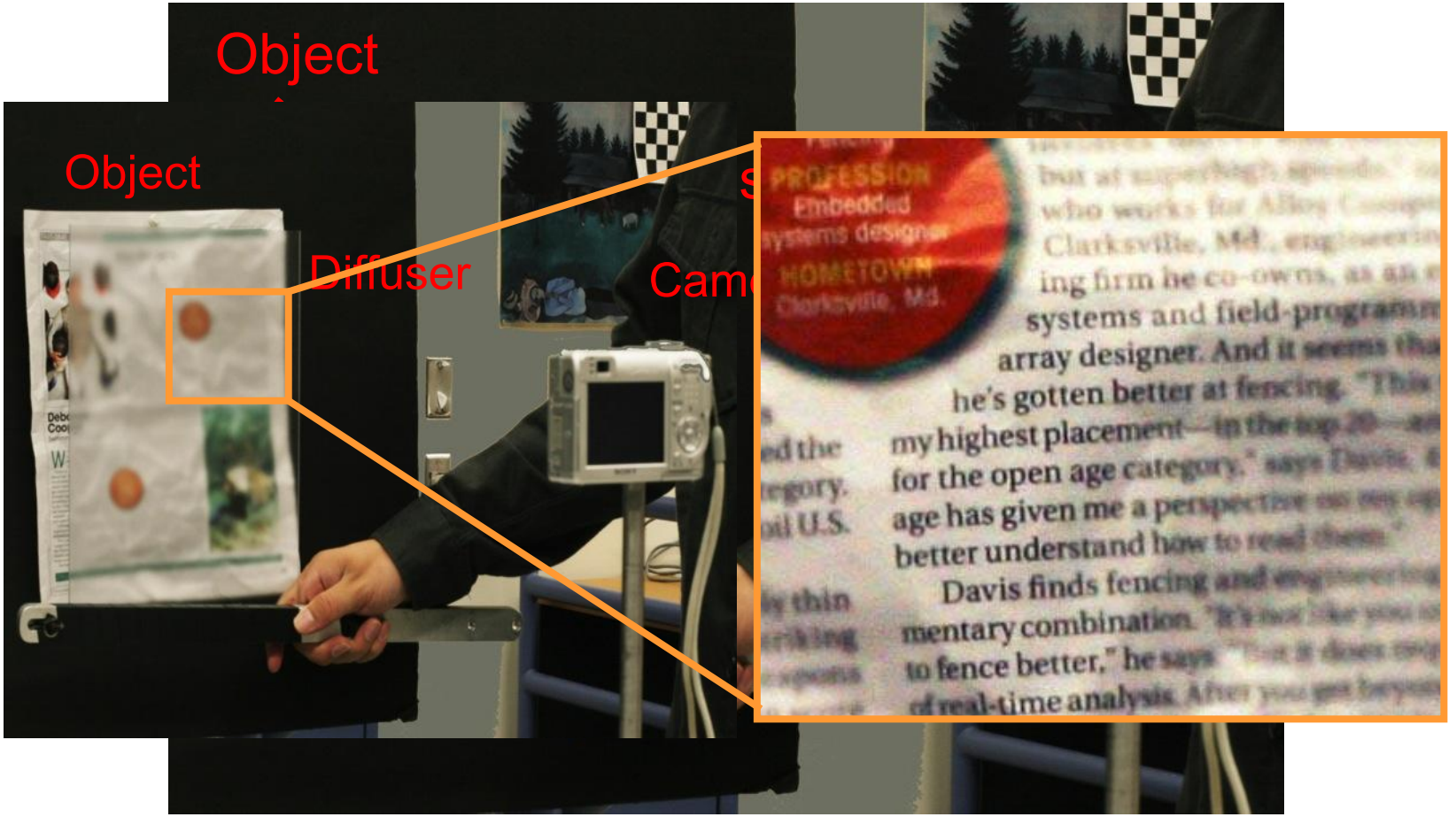


Diffuser to
preview the image



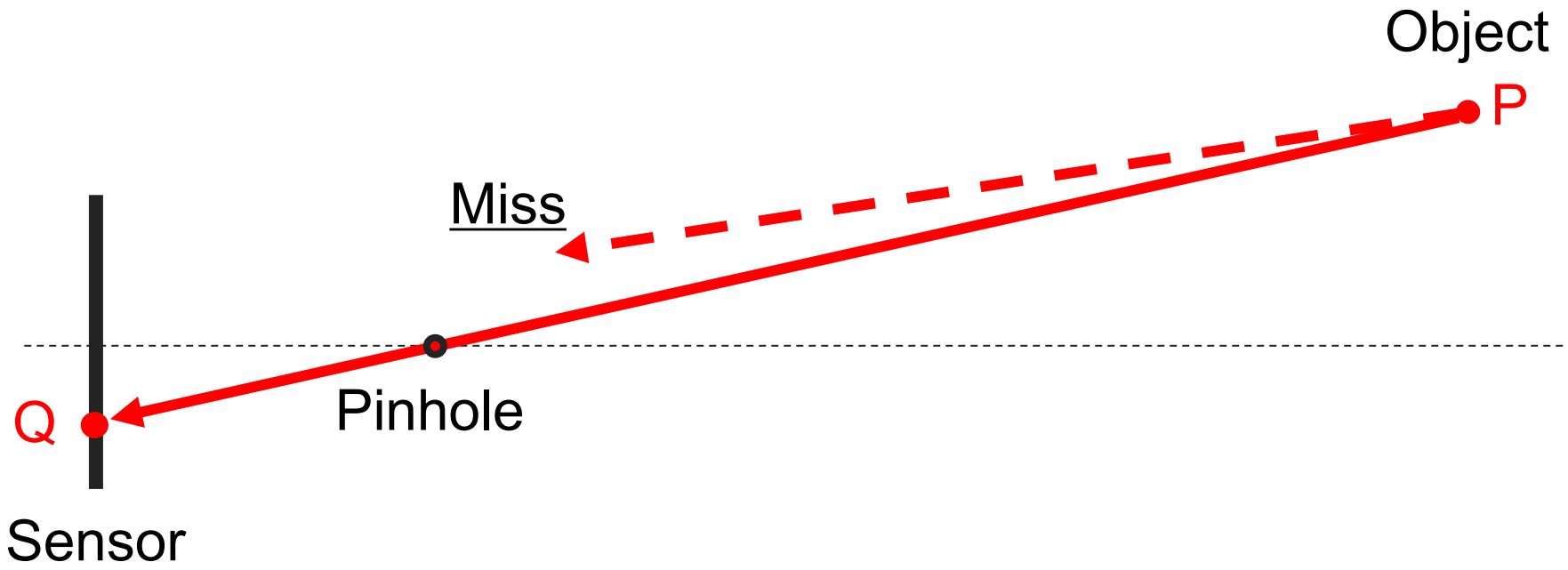
(B&H)
Diffusers to soften the image

Diffusion Encodes Depth

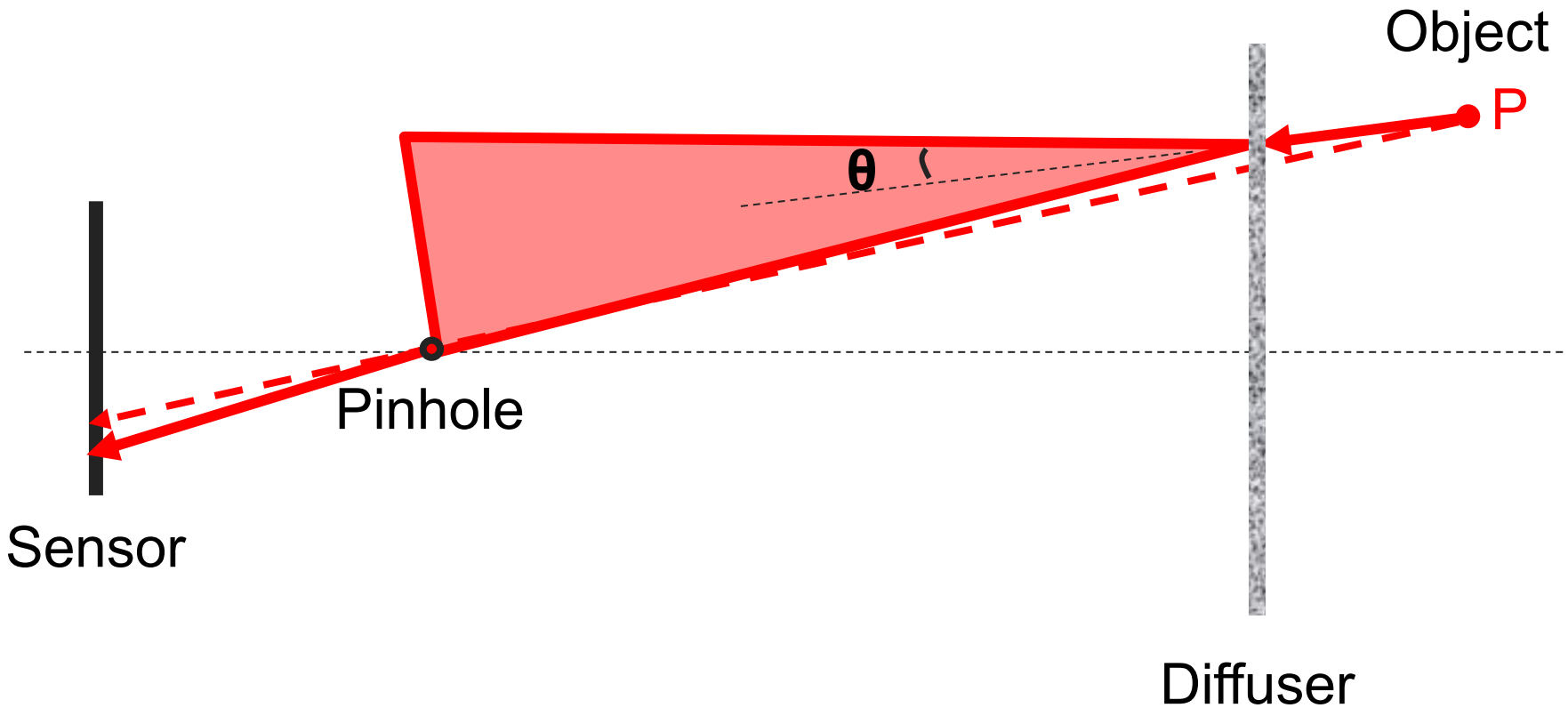


The amount of diffusion varies with depth.

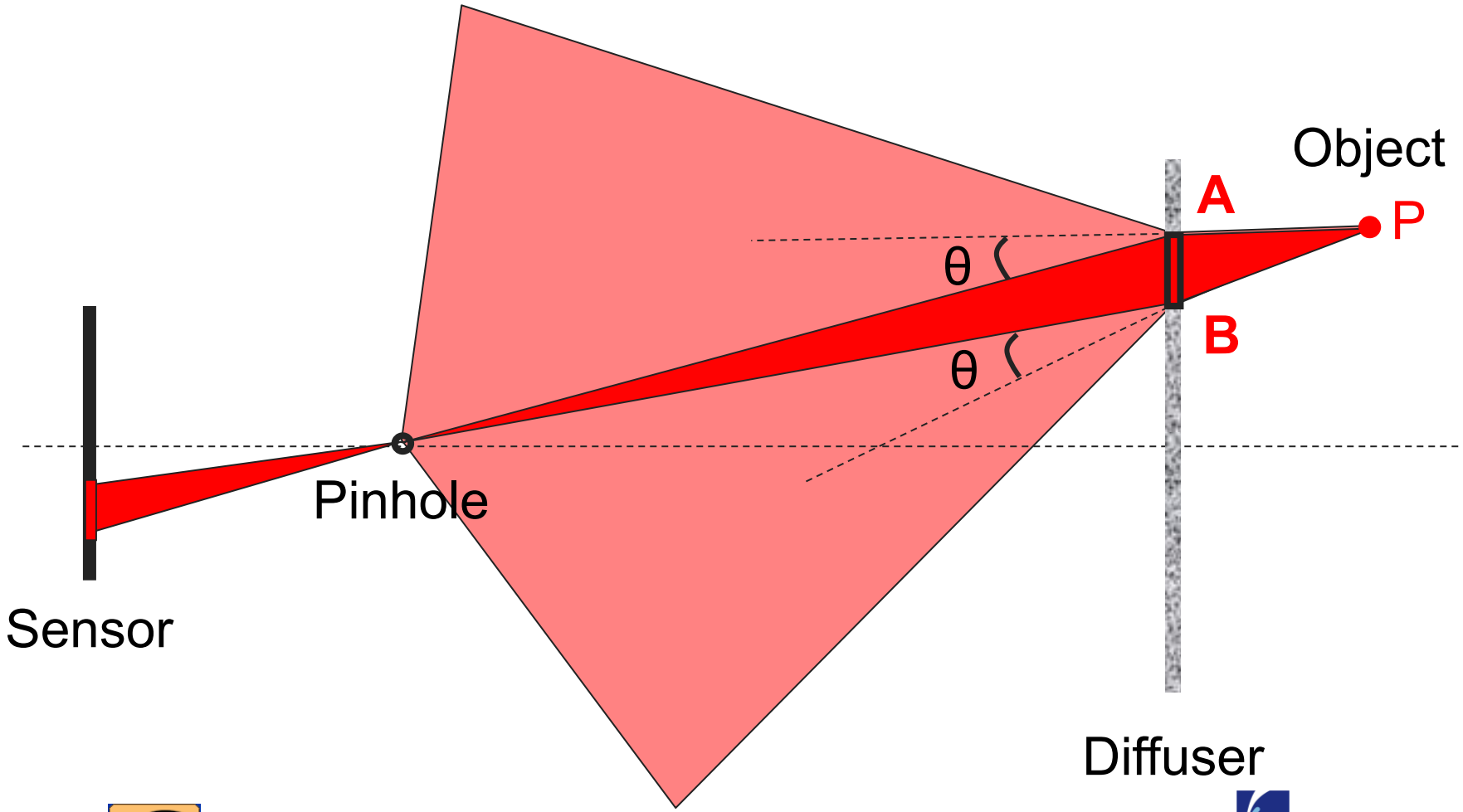
Geometry of Diffusion: A Pinhole Camera



Geometry of Diffusion: A Pinhole Camera



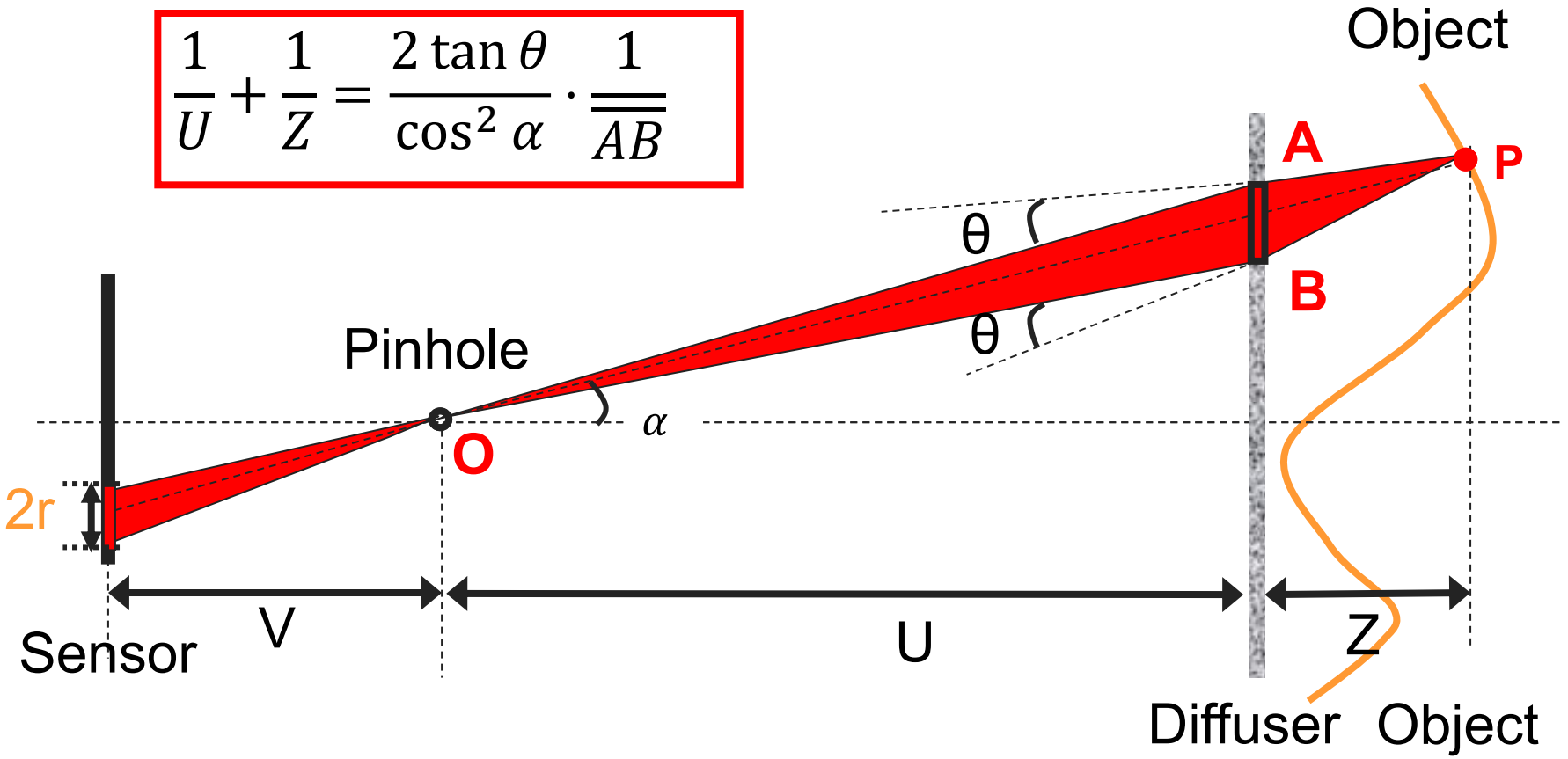
Geometry of Diffusion: A Pinhole Camera



Geometry of Diffusion: A Pinhole Camera

Diffusion Law:

$$\frac{1}{U}+\frac{1}{Z}=\frac{2\tan\theta}{\cos^2\alpha}\cdot\frac{1}{\overline{AB}}$$

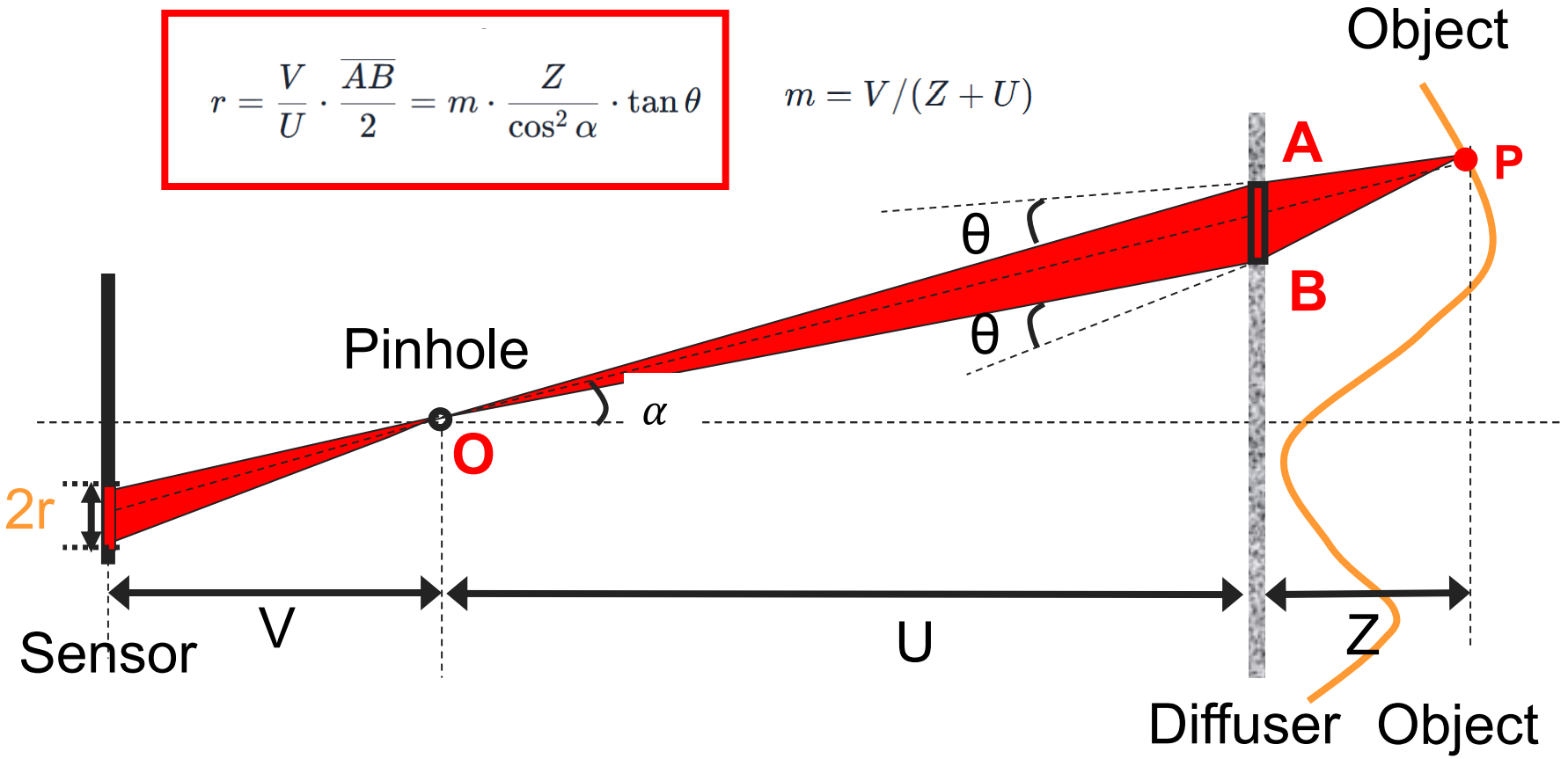


Geometry of Diffusion: A Pinhole Camera

Diffusion Size and Depth:

$$r = \frac{V}{U} \cdot \frac{\overline{AB}}{2} = m \cdot \frac{Z}{\cos^2 \alpha} \cdot \tan\theta$$

$$m = V/(Z+U)$$

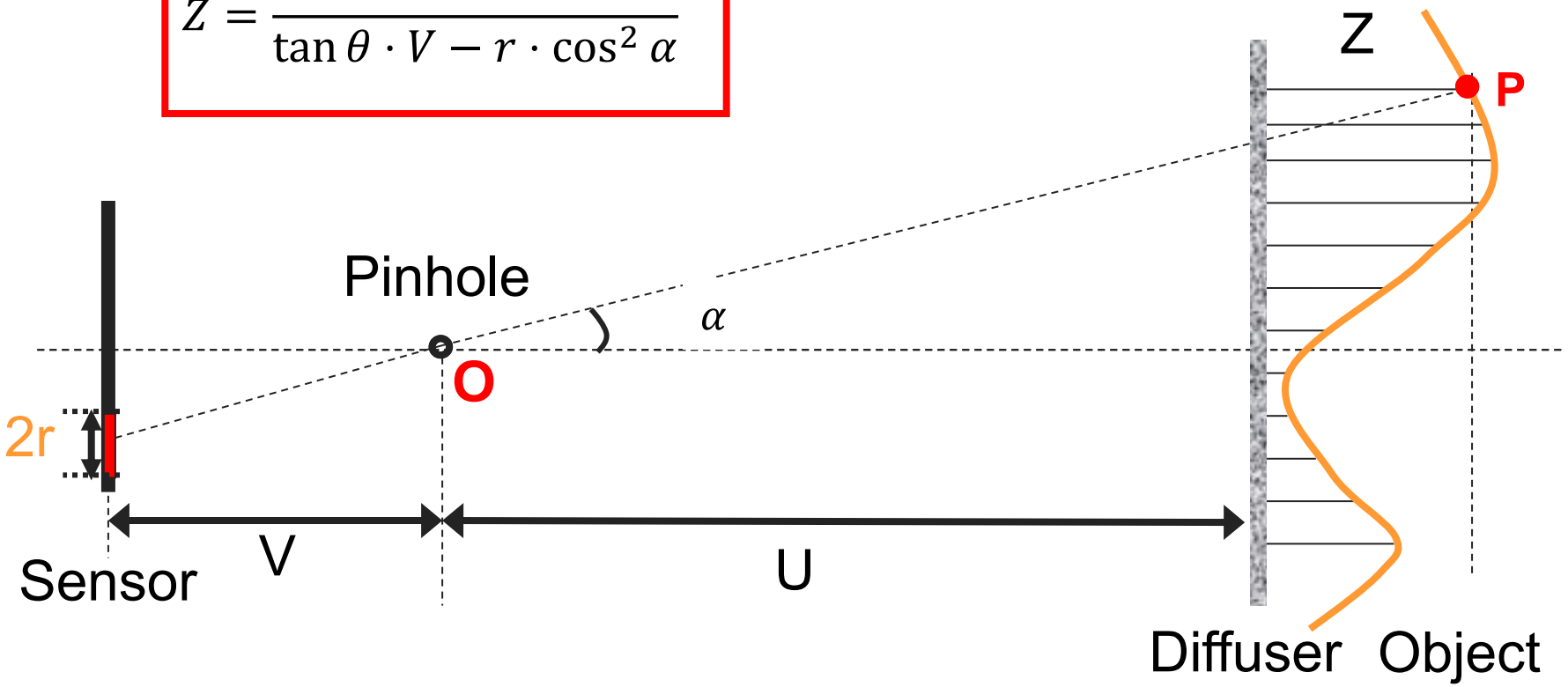


Geometry of Diffusion: A Pinhole Camera

Diffusion Size and Depth:

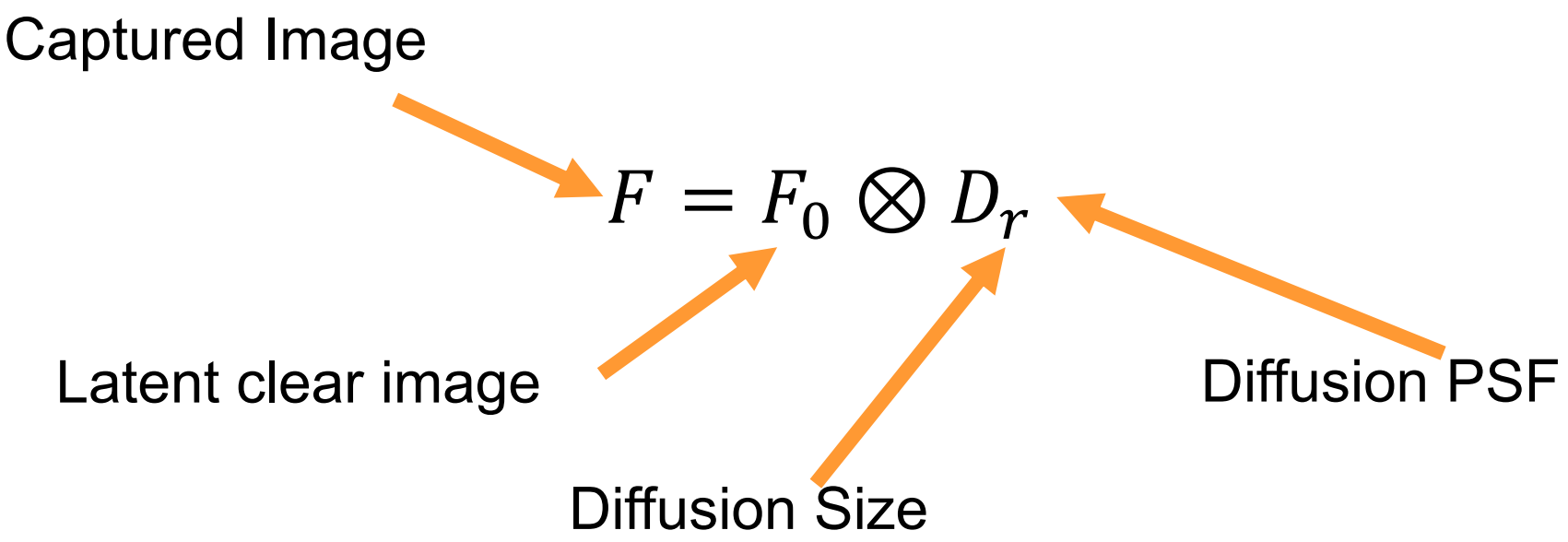
$$Z = \frac{r \cdot \cos^2 \alpha \cdot U}{\tan \theta \cdot V - r \cdot \cos^2 \alpha}$$

Diffuser as a proxy object



Diffusion as Convolution: A Pinhole Camera

Assume field angle and depth are constant for small image patches, we have:



Experiments



Five playing cards,
0.29mm thick each

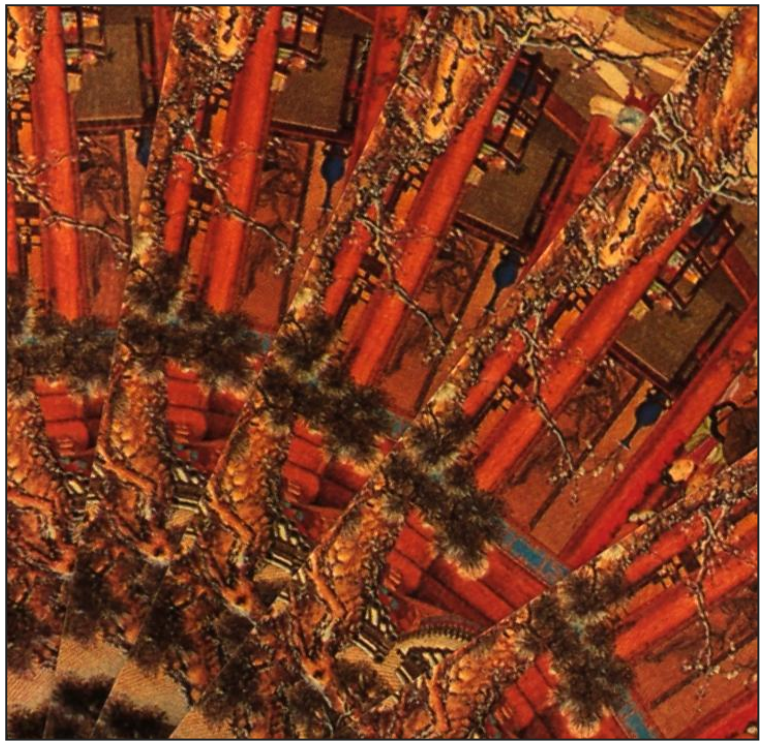


Canon 20D + 50mm Lens

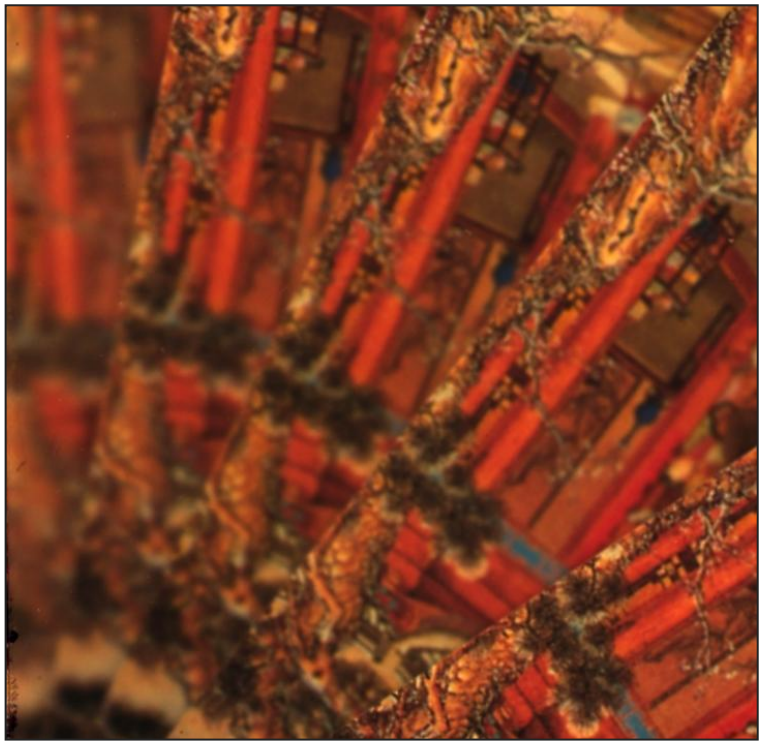


Luminit Diffuser (20°)

Experiments



Captured WITHOUT a Diffuser

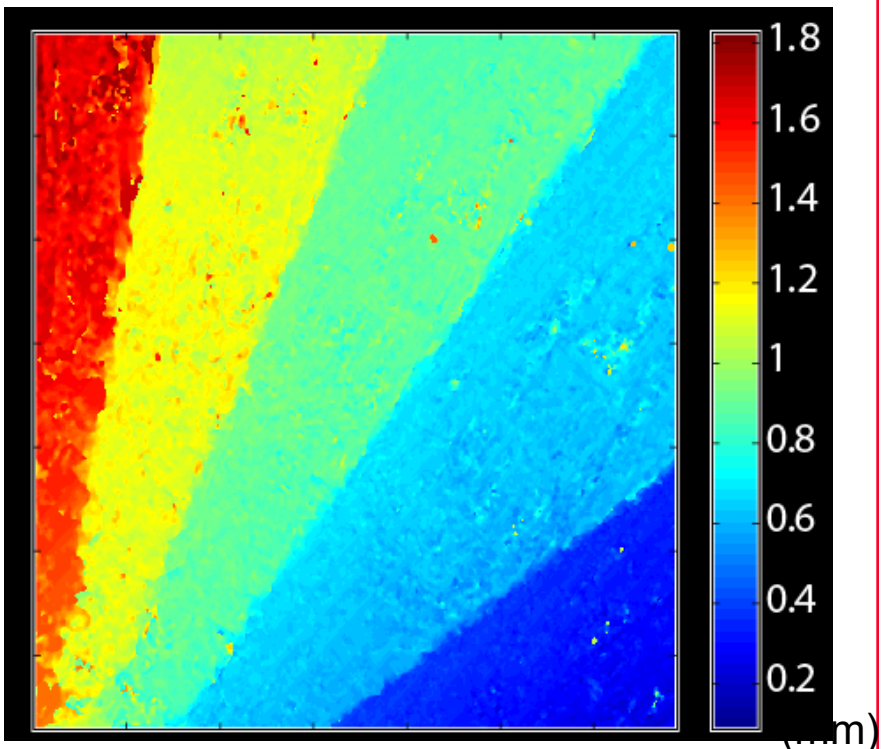


Captured WITH a Diffuser

Experiments



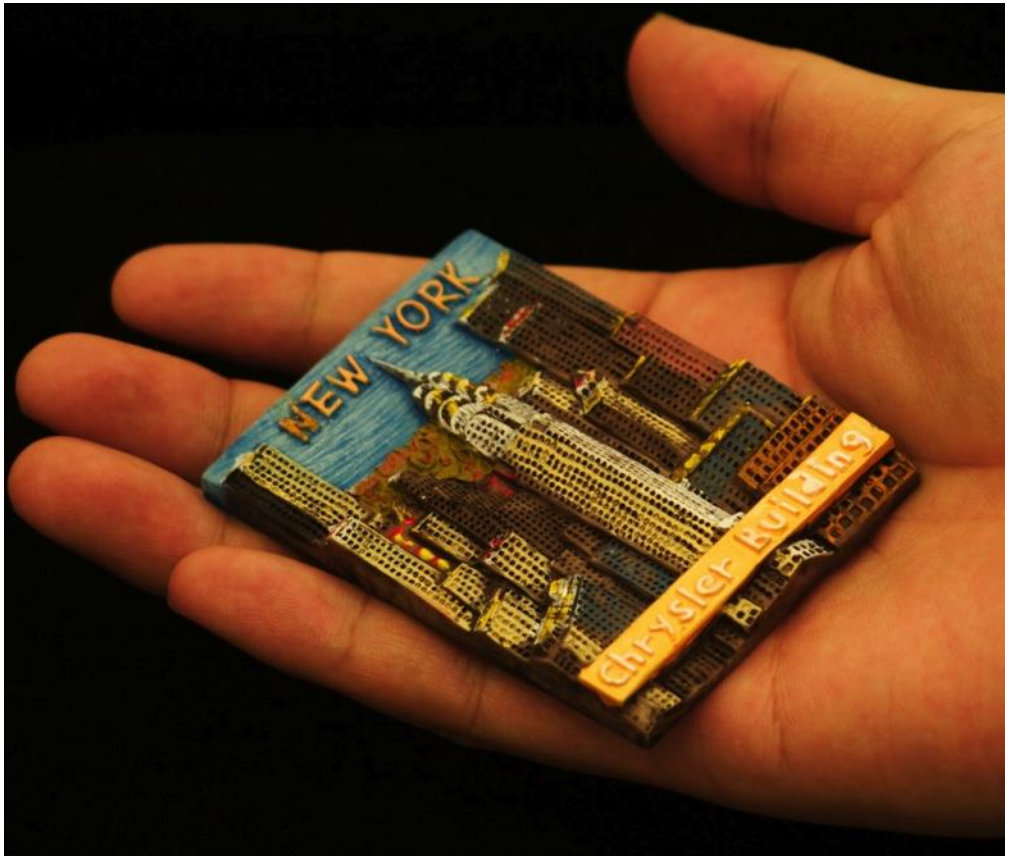
Five playing cards,
0.29mm thick each



Computed Depth Map
(~ 0.1 mm precision)

Experiments

A small sculpture of
about 4mm thickness





Canon G5
Compact Camera



Luminit Diffuser (5°)

Experiments



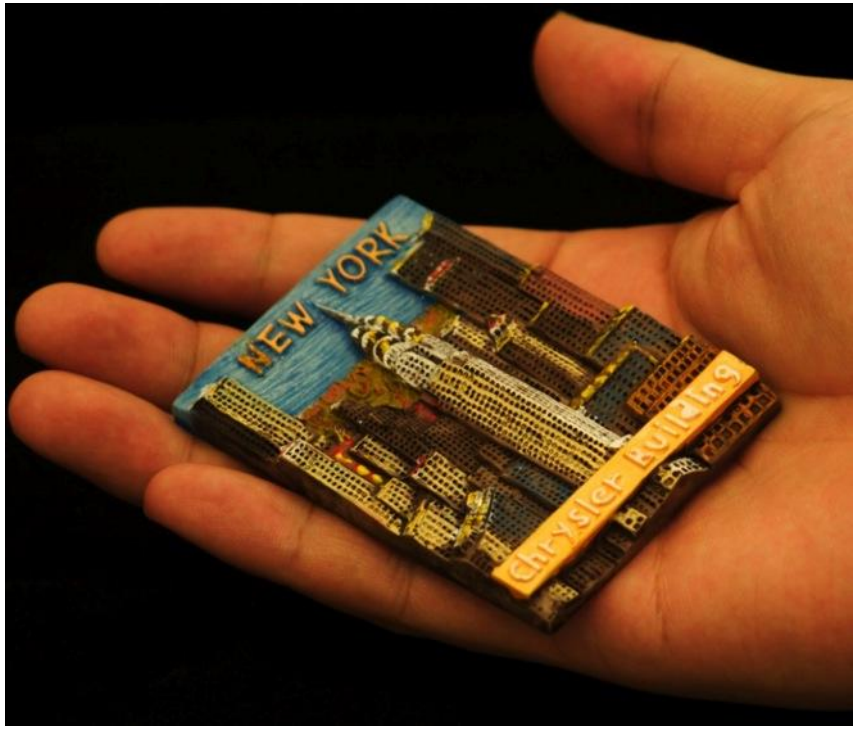
Captured WITHOUT a Diffuser

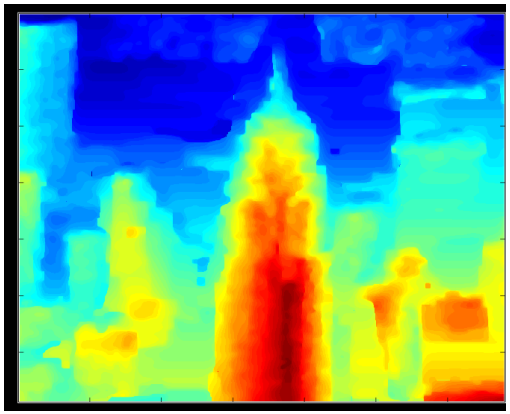


Captured WITH a Diffuser

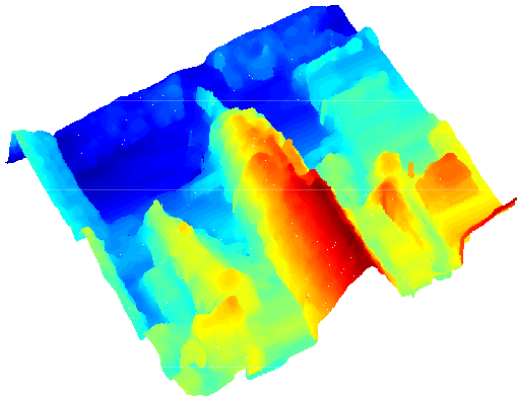
Experiments

A small sculpture of about 4*mm* thickness



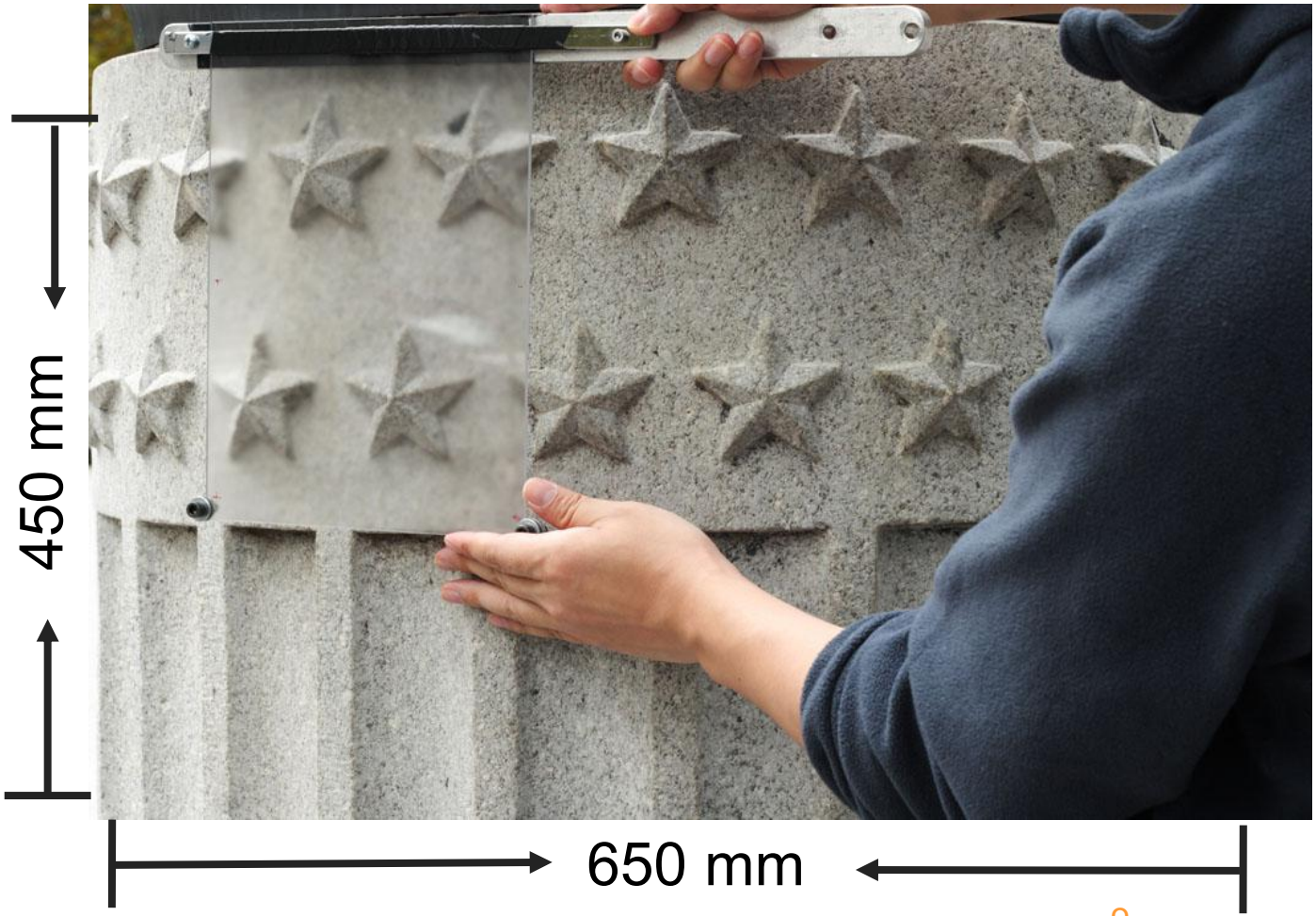


Computed Depth Map



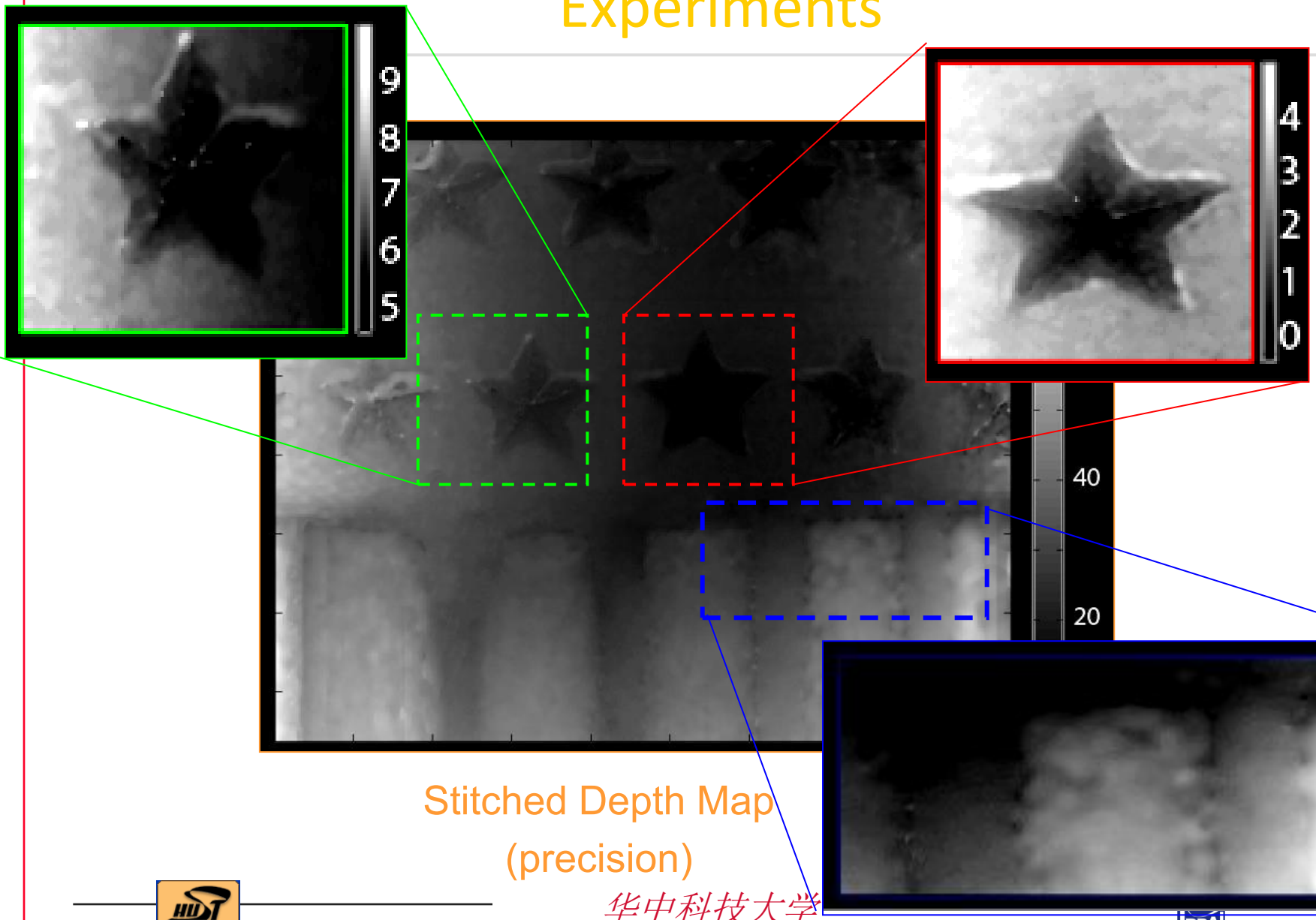
A 3D View of Depth Map

Experiments



Canon 20D; Gaussian Diffuser (10°)

Experiments



Stitched Depth Map

(precision)

The end !